

Proximity to the Plague: Measuring Locust Exposure and Child Health*

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Abstract

I investigate the early childhood health outcomes of the Kenyan 2020–2021 desert locust invasion. To account for the multi harvest setting, I build a treatment that conditions on the insect’s biological capacity to damage vegetation, the growth stage of the local crop, and subsequent vegetal decline, matching geo-referenced locust observations to the 2022 Kenya Demographic and Health Survey. Children exposed in utero within 30 km of a verified damage event carry weight-for-height z-scores 0.185 standard deviations below the control group, and are 26 percent more likely to be clinically wasted. Height-for-age does not move. The estimate loads on third-trimester exposure. The effect is largest in riverine livelihood zones, where households depend most heavily on their own harvest and have the least market cushion when it fails: exposed riverine children lose an additional 0.25 standard deviations on weight-for-height beyond the average cropping-zone effect.

Keywords: desert locust, child health, wasting, exposure measurement, Kenya

JEL Codes: I15, O12, Q12, Q54, C18

*All errors are my own.

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1 Introduction

Cross-sectional household surveys such as the Demographic and Health Surveys (DHS) and the Multiple Indicator Cluster Surveys (MICS) have become a workhorse for identifying the consequences of natural disasters in cross-sectional microdata (Michalopoulos and Papaioannou, 2018). As these surveys provide geocoded sample clusters, a researcher can link a household to the shock of interest by drawing a radius around the observed event and treating everyone inside this radial “buffer” as exposed.

A small but growing literature in economics has come to employ these techniques to the incidence of locust plagues (Chatterjee, 2022; Le and Nguyen, 2022; De Vrejer et al., 2015). These papers often assign exposure from a locust’s recorded presence within a radius or an administrative boundary, without conditioning on the insect’s biological capacity to harm vegetation, the phase of a crop’s life cycle or on whether a crop was subsequently damaged following exposure to the plagues. Conte et al. (2023) use a proximity-based measure and rely anecdotally on Mali’s primarily single harvest to time specific pathways to health from the locusts. However, many countries hit by these types of agricultural shocks are bimodal; their harvests follow from two rainy seasons in a year. Indeed, the 2020-2021 shock this paper studies primarily affected bimodal East African countries such as Uganda, Ethiopia and Kenya. This paper attempts to accurately assess causal health outcomes in a bimodal multi-crop setting using the 2020-2021 desert locust invasion in Kenya.

I match georeferenced locust observations to the 2022 Kenya Demographic and Health Survey and build an exposure measure that requires the insect to have arrived when a crop was actually standing in the field. Kenya’s bimodal rainfall gives it two planting windows, two harvests, and multiple crop calendars that differ across the 24 livelihood zones by several months, so the same calendar date maps to a susceptible crop in one zone and bare ground in another. I use that calendar to gate exposure: a locust record counts only if it falls within a given radius of a survey cluster while the child was in utero, if it describes a gregarious form capable of stripping a field, if the local crop calendar places a crop in a susceptible growth stage on the record’s date, and if satellite vegetation over the cluster then declines relative to that cluster’s own pre-invasion climatology. I build the measure at nine radii from 10 to 50 km and fix 30 km in advance as the main radius.

Children exposed in utero under this measure carry weight-for-height z-scores 0.185 standard deviations below unexposed children in the same survey cluster born in a different month. Adding county-specific linear time trends raises the estimate to 0.205. At the clinical threshold the probability of wasting rises 1.9 percentage points against a baseline of 7.2 percent, a relative increase of 26 percent, and 3.6 percentage points once county trends are absorbed. The estimate is stable across the child, family, and trend specifications and is significant at the 1 percent level throughout. It holds at every radius from 10 to 50 km and grows as the radius tightens, reaching 0.312 standard deviations within 10 km.

Wasting records a deficit of weight relative to attained height and responds to adverse nutritional effects in the short term within weeks. Height-for-age records accumulated growth over the whole of a child’s life and moves slowly. The median child in this sample was 14 months old at measurement, so a disruption confined to gestation and the months around birth has had little time to register in linear growth. The estimates match that prediction. No height-for-age coefficient is significant at the main radius, and the effect loads on third-trimester exposure, the window Stein and Susser (1975) identify as carrying the whole of the Dutch famine’s effect on birth weight while leaving length at birth nearly untouched.

In addition to the novel empirical strategy, this work contributes to the fetal origins literature. Barker (1990) proposed that adult disease traces to conditions in-utero, and economists have since documented effects of prenatal shocks on long-term health, schooling, and earnings (Almond and Currie, 2011; Almond

et al., 2018). Rainfall and drought supply most of the identifying variation. Maccini and Yang (2009) link early-life rainfall to adult height and schooling in Indonesia, Hoddinott and Kinsey (2001a) and Alderman et al. (2006) document persistent growth faltering after Zimbabwe’s droughts, and Shah and Steinberg (2017) show that rainfall raises human capital in utero while lowering it at school age. Anthropometric z-scores are the standard outcome because early childhood growth predicts adult human capital (Victora et al., 2008). Rainfall shocks identify the effect of income and food availability on the fetus. But they cannot separate the agricultural channel from the waterborne-disease channel, the maternal-labor channel, or the price channel. This paper isolates the agricultural channel by construction: the shock destroys crops and nothing else and identifies the effect of this acute vegetal damage on early childhood health.

The remainder of this paper is organized as follows. Section 2 describes the locust, Kenya’s agricultural landscape, and the 2020–2021 invasion. Section 3 describes the data. Section 4 builds the exposure measure and the estimating equation. Section 5 reports results and robustness. Section 5.4 examines mechanisms. Section 6 concludes.

2 Setting

2.1 The Desert Locust

The desert locust, *Schistocerca gregaria*, is a grasshopper that changes phase with population density. At low densities it is solitary (solitarious). These locusts avoid one another, move relatively little, and typically remain within a restricted area, dispersing rather than congregating. Their coloration is usually cryptic, matching the background vegetation and soil, and their bodies are slimmer with proportionally longer legs. Because they neither aggregate nor migrate in large numbers, solitary locusts cause little or no damage to crops and are not considered a plague threat. This phase persists as long as densities remain low, and it is the state in which locust populations spend most of their time.

In the gregarious phase, by contrast, locusts are crowded and highly social. When rainfall concentrates individuals onto scarce green vegetation, crowding triggers a transformation toward this phase over hours to generations (Symmons and Cressman, 2001). Gregarious individuals aggregate, march as hopper bands before they can fly, and form adult swarms that travel with the wind over hundreds of kilometers. They develop striking black-and-yellow (or pink in immature adults) warning coloration, broader bodies, and shorter legs, and they are more active and voracious than their solitary counterparts. Swarms can contain billions of individuals and consume vast quantities of vegetation in a single day, making the gregarious phase the one responsible for the devastating agricultural damage for which desert locusts are notorious. The two phases are not fixed states but endpoints of a continuous, density-dependent continuum, with a transient phase occupying the middle ground.

Only gregarious swarms and hopper bands reach the densities that strip a field. Solitarious individuals occur throughout the locust’s recession area in most years without agronomic consequence.

Damage also depends on timing. Locusts are polyphagous and consume cereals, legumes, and pasture, but the loss a field sustains depends on what stage the crop has reached. Standing biomass close to harvest cannot be replaced within a season. A field at planting can be resown at the cost of seed and labor. Grain already harvested and stored is not at risk, and a field out of season has nothing to lose. A swarm passing over a harvested field is agronomically irrelevant even though it appears in the record as a locust observation.

Upsurges are episodic. Global control efforts suppressed them for much of the late twentieth century

(Zhang et al., 2019), and recent work links renewed activity to extreme rainfall events and their climatic drivers (Salih et al., 2020; Liu et al., 2024; Peng et al., 2020).

2.2 Kenya’s Agricultural Landscape

Kenya is a country located on the eastern coast of Africa, lying directly across the Equator. Its capital city is Nairobi, which serves as the nation’s commercial and diplomatic hub. Administratively, Kenya is divided into 47 semi-autonomous regions known as counties, a system established under the 2010 Constitution. On the map, Kenya shares borders with Tanzania to the south, Uganda to the west, South Sudan to the northwest, Ethiopia to the north, and Somalia to the northeast, while its southeastern edge meets the Indian Ocean (Ntarangwi et al., 2026).

Kenyan agriculture is organized around two rainy seasons. The long rains fall from roughly March to May and the short rains from October to December. Most of the country’s maize, the dominant food staple, is planted with the long rains in the western and Rift Valley highlands and harvested from August onward. Eastern and coastal areas depend more heavily on the short rains and on drought-tolerant crops including sorghum, millet, and cowpeas. The timing of planting, vegetative growth, and harvest therefore differs across the country by several months, and a locust arrival that is catastrophic in one county may be harmless in another on the same date.

County boundaries do not track how households feed themselves. Figure 1 maps the FEWS NET 24 livelihood zones under the household economy approach, each covering an area whose residents share a production system and a set of food-access strategies (Famine Early Warning Systems Network (FEWS NET), 2011).

Cropping zones cover the western and central highlands, the eastern slope of the highlands, and the coastal strip. They include the Central Highlands High Potential Zone (KE19), the Western Medium and High Potential Zones (KE20 and KE21), the Southeastern mixed farming zones (KE16 and KE17), and the two coastal farming zones (KE12 and KE13). Households in these zones grow maize, beans, potatoes, and in the highlands tea and coffee, and buy the balance of their food.

Pastoral zones cover the arid north and northeast and a belt of southern rangeland. They include the Northwestern, Northern, Northeastern, Eastern, Southeastern, and Southern Pastoral Zones (KE01, KE05, KE09, KE10, KE11, KE15). Households herd camels, cattle, and small stock, move with the rains, and obtain food from milk, livestock sales, and purchase.

Agropastoral zones sit on the transition between the two and combine rain-fed cropping with herding (KE03, KE07, KE18, KE24). They hold 986 children. Riverine zones are narrow irrigated strips along the Turkwell, the Dawa at Mandera, and the Tana (KE02, KE08, KE14), holding 196 children. Fishing zones cover the shores of Lake Turkana and Lake Victoria (KE04, KE23).

The distinction matters for measurement. Locusts consume pasture as well as crops (Le Gall et al., 2019), so pastoral households are not unaffected, but the mechanism and the timing differ from crop loss. A crop calendar governs when a field is vulnerable, and a zone with no crops has no such calendar. The FEWS NET seasonal calendar for a pastoral zone contains no crop row at all, only livestock production, migration, and market access. An exposure measure gated on crop growth stage therefore cannot flag a pastoral household, and a measure gated on proximity alone flags them at the same rate as anyone else.

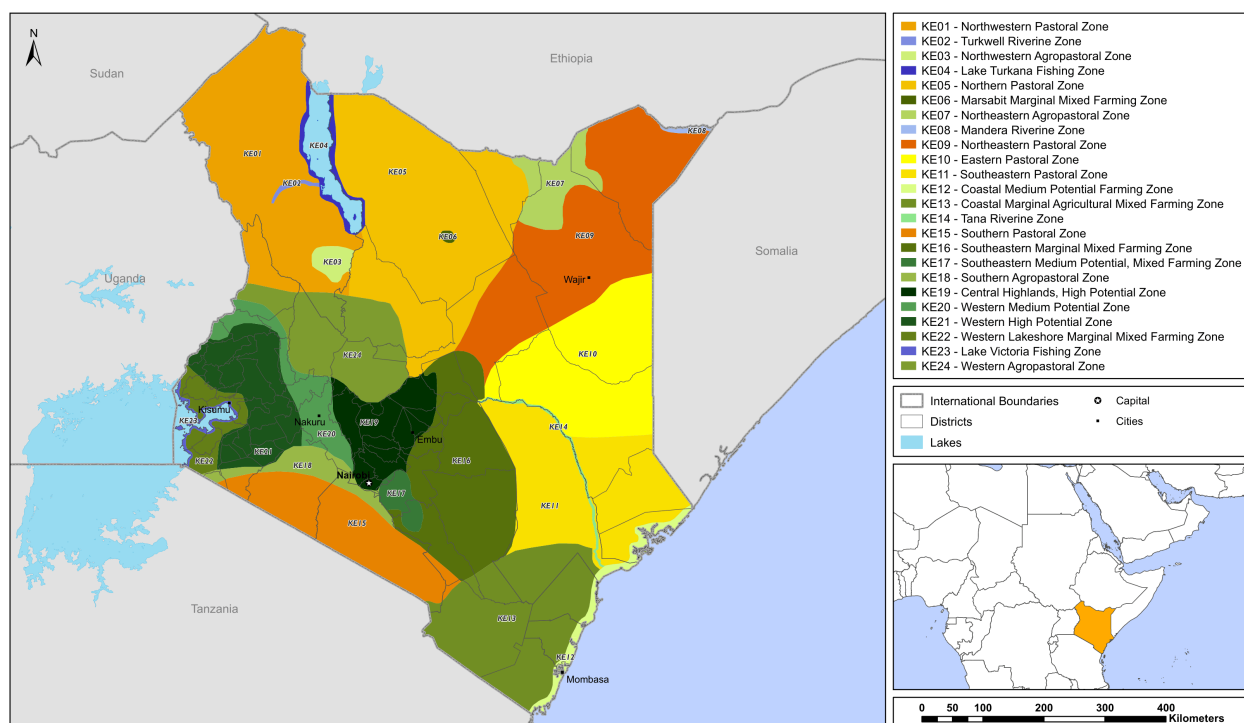


Figure 1: FEWS NET livelihood zones of Kenya. Source: (Famine Early Warning Systems Network (FEWS NET), 2011)

2.3 The 2020–2021 Kenyan Crisis

Heavy and repeated rainfall over the Arabian Peninsula in 2018 and 2019 produced several undetected breeding generations of the desert locust. Tropical cyclones delivered exceptional precipitation to usually arid terrain, creating temporary lakes in the desert and allowing locusts to breed (Salih et al., 2020). These favorable conditions sustained three generations of breeding in roughly nine months, roughly a twentyfold increase over a typical year.

Swarms then crossed into the Horn of Africa through Yemen, Somalia, and Ethiopia, reaching northern Kenya in late December 2019. By early 2020 swarms had spread across much of central and eastern Kenya, prompting the national government to declare an unprecedented threat to food security and livelihoods. The invasion represented the worst locust upsurge Kenya had experienced in seventy years.

The upsurge continued into 2021. A second wave entered Kenya from Ethiopia and Somalia in mid-November 2020 and eventually reached 15 counties, reversing the gains of the preceding months (Ministry of Agriculture, Livestock, Fisheries and Cooperatives, Kenya, 2021). Kenya responded with nine surveillance and spray aircraft, 21 ground spray vehicles, and more than 500 trained personnel, operating alongside FAO-supported control operations (EnviroNews Nigeria, 2021). Ground and aerial spraying proceeded while breeding continued, and control teams worked to suppress new hopper bands before they could fledge into swarms. The campaign brought the invasion largely under control by mid 2021 with some gregarious sightings in the fall of that year. The FAO declared the invasion officially over in February 2022 (Food and Agriculture Organization of the United Nations, 2021).

Figure 2 plots daily counts of gregarious swarm and hopper-band observations across the episode. The invasion arrives in two waves. The first builds through January and February 2020, peaks in May and June,

and subsides by August. The second builds from December 2020, peaks in February 2021, and ends in April. The shock therefore ran for sixteen months and crossed three long-rains seasons and two short-rains seasons.

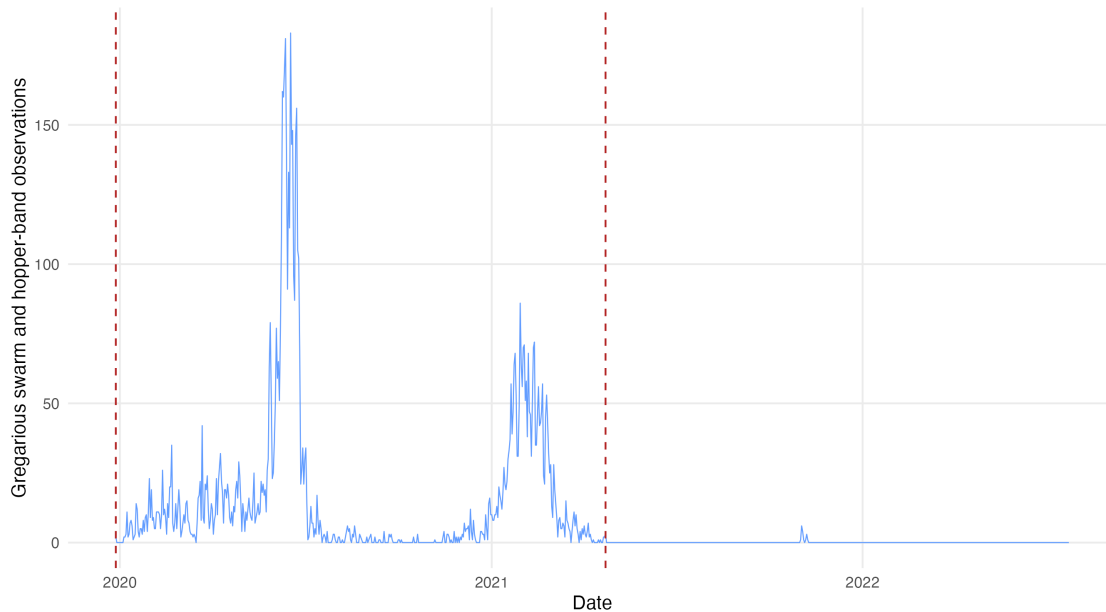


Figure 2: Daily gregarious swarm and hopper-band observations in Kenya. Dashed lines mark the derived invasion window.

3 Data

3.1 Household Survey

The child-level sample is drawn from the 2022 Kenya Demographic and Health Survey (DHS), with the 2014 round used as a pre-invasion falsification sample ([Kenya National Bureau of Statistics \(KNBS\) and ICF, 2023](#); [Kenya National Bureau of Statistics \(KNBS\) and ICF International, 2015](#)). The DHS is a nationally representative household survey conducted under the DHS Program (ICF, formally funded principally by USAID), fielded in Kenya roughly every five years. Each round enumerates all children born to sampled women in the preceding five years, with retrospective birth histories, anthropometric measurement (height and weight, converted to age- and sex-standardized z-scores against the WHO growth reference of [WHO Multicentre Growth Reference Study Group \(2006\)](#)), and household, maternal, and child covariates.

Each DHS cluster (an enumeration area, roughly a village or urban block) is assigned a single GPS coordinate, displaced by up to 2 km for urban clusters and 5 km for rural clusters (up to 10 km for 1 percent of rural clusters, restricted to stay within the same second administrative region) to protect respondent confidentiality ([Burgert et al., 2013](#)). Raster and vector extraction in this paper such as rainfall and distance to roads, is performed as a buffer mean or nearest-feature match over this displacement radius rather than at the point coordinate.

The 7,603 children with valid anthropometry are spread across 1,644 clusters, a mean of 4.6 children per cluster and a median of 4, ranging from a single measured child in the thinnest clusters to 19 in the largest.

3.2 Locust Records

Locust observation records come from the Reconnaissance and Management System of the Environment of *Schistocerca* (RAMSES), the geographic information system used by national Locust Information Officers in each affected country to record survey and control activity, and consolidated by the Food and Agriculture Organization’s Desert Locust Information Service (DLIS) ([Food and Agriculture Organization of the United Nations \(FAO\), 2022](#)). Records are distributed through the FAO Locust Hub. Each record carries a date, coordinates, a presence type (e.g. hopper band, adult swarm, or an explicit absence), a phase classification (gregarious, transiens, or solitary), a maturity classification, and, where control operations occurred, the control type.¹

3.3 Climate and Vegetation

I use three remotely sensed climate products; rainfall and two measures of vegetal health. Rainfall comes from the Climate Hazards group InfraRed Precipitation with Station data (CHIRPS), a quasi-global, 0.05°-resolution, daily-to-monthly precipitation dataset combining infrared satellite imagery with station observations, produced by the Climate Hazards Center at UC Santa Barbara and designed for drought monitoring in data-sparse regions ([Funk et al., 2015](#)). Both monthly and dekadal (10-day) CHIRPS products are used: the monthly product supports gestational rainfall controls and the drought construction described next, and the dekadal product supports growing-season onset detection for the crop calendar.

The Standardized Precipitation Evapotranspiration Index (SPEI), at 6-month accumulation windows, is constructed from CHIRPS precipitation and potential evapotranspiration (PET) from TerraClimate, a global, ~4 km monthly climate and water-balance dataset ([Abatzoglou et al., 2018](#)). SPEI standardises the climatic water balance (precipitation minus PET) rather than precipitation alone, following the index definition in [Vicente-Serrano et al. \(2010\)](#); because the drought conditions overlapping the invasion period combined rainfall deficits with elevated evaporative demand, a precipitation-only index might understate its effects.

Vegetation condition comes from MODIS/Terra Vegetation Indices (MOD13Q1), a 16-day, 250 m composite Normalized Difference Vegetation Index (NDVI) product from NASA’s MODIS sensor, accessed via the AppEEARS extraction service ([Didan, 2015](#)). Pixels are screened on the product’s reliability flag before averaging, so cloud- or snow-contaminated observations never enter a cluster mean. NDVI levels are converted to per-cluster, per-composite z-scores against a pre-invasion climatology.

3.4 Crop Calendars

The exposure measure requires knowing, for a given cluster on a given date, whether a crop was standing in a field and how far it had advanced. Livelihood zone boundaries come from shape files that FEWS NET provides; the FEWS NET Kenya livelihood zone map described in Section 2 ([Famine Early Warning Systems Network \(FEWS NET\), 2011](#)).

The FEWS NET livelihood profiles also published a seasonal calendar for each of Kenya’s 24 livelihood zones, showing land preparation, planting, and harvesting by crop alongside livestock, labor, and market-access cycles ([Famine Early Warning Systems Network \(FEWS NET\), 2011](#)). A cropping zone’s calendar lists maize, beans, potatoes, and the relevant cash crops. A pastoral zone’s calendar lists calving, kidding,

¹Unfortunately, control operation data are sparse and recorded with many errors in this sample.

migration, and milk availability, and contains no crop row. Clusters in zones without a crop row are never placed in a susceptible stage, so they can never satisfy the stage condition.

Each seasonal calendar with a three-color Gantt legend, land preparation, planting, and harvest, plotted month by month for every crop grown in that zone ([Famine Early Warning Systems Network \(FEWS NET\), 2011](#)).² An automated pixel-extraction routine locates the table grid and the legend’s three reference colors on each of the 24 zone pages, then classifies every crop row’s bar by nearest-color match at semi-month resolution. Seventeen zones return at least one crop row. The other seven are auto-detected as having no crop calendar at all, and these seven are exactly the pastoral and fishing zones of Section 2.³ The automated output was checked against an independent manual pixel read of one page, which caught and fixed an off-by-one error in which row’s bar was attributed to which crop, and every zone runs through the identical code path after that fix.

The three-color legend collapses to a single susceptibility ranking. Land preparation leaves bare ground and is not susceptible. Planting is moderately susceptible, since a destroyed planting can be resown at the cost of seed and labor. Harvest, which on these pages denotes the full standing-crop period from establishment through the actual harvest rather than the narrow act of reaping, is the most susceptible stage. A cluster-date outside every crop’s bar on its zone’s page is out of season.

Where a cluster has more than one crop group or more than one season active on a date, the governing stage is the most susceptible of those active, as Section 4 formalizes this.

3.5 Food Prices

Prices come from the World Bank’s Real-Time Prices (RTP) platform, which supplies a monthly price estimate for maize and beans at 234 Kenyan markets from 2007 onward, combining sparse directly-observed transactions with a model that imputes missing values from nearby markets, related commodities, and historical patterns, following the method in [Andrée \(2021\)](#). I take the platform’s own “close” estimate, its best and most current price for a given market-month, as given rather than re-deriving it; I restrict the sample to 2015–2022, the period spanning the children in the main analysis.

3.6 Roads

Distance from each DHS cluster to the nearest major road is computed from OpenStreetMap’s Kenya road network, exported through the Humanitarian OpenStreetMap Team’s export tool ([Humanitarian OpenStreetMap Team \(HOT\) and OpenStreetMap contributors, 2022](#)). This is a crowd-sourced, continuously updated vector road network.

3.7 Administrative Boundaries

County boundaries come from the geoBoundaries Global Database of Political Administrative Boundaries, distributed via the Humanitarian Data Exchange (HDX) as the Kenya subnational administrative boundaries dataset. Kenya’s 47 counties correspond to the ADM1 level in this dataset, which is produced and maintained by geoBoundaries and published under an open license ([geoBoundaries, 2022](#)).

²Figures 12 and 13 in Appendix B provide examples of one calendar for each type.

³Three zones, Tana Riverine, Western High Potential, and Lake Victoria Fishing, needed the corresponding page read by hand instead: the automated color-distance threshold that separates the legend colors was not stable across the environments the pipeline was run in for these three pages specifically, so rather than tune a threshold, those three pages are transcribed directly and flagged as such in the underlying data.

4 Empirical Strategy

4.1 Exposure construction

Let $\mathcal{L}$ denote the set of locust monitoring records, where each record $\ell \in \mathcal{L}$ carries a location, an observation date δ_ℓ , a presence indicator $P_\ell \in \{0, 1\}$, and a damage classification $\kappa_\ell \in \{\text{low, moderate, high, solitary, none}\}$ assigned from the record's presence type and dominant maturity. A record's *form* is

$$\text{Form}(\ell) = \begin{cases} \text{damaging} & \text{if } P_\ell = 1 \text{ and } \kappa_\ell \in \{\text{low, moderate, high}\} \\ \text{benign} & \text{if } P_\ell = 1 \text{ and } \kappa_\ell = \text{solitary} \\ \text{absent} & \text{if } P_\ell = 0. \end{cases} \quad (1)$$

Damaging records are gregarious swarms and hopper bands. Benign records are solitary or transients sightings, retained throughout as the falsification group in Section 5.3.

For child i born at date b_i to survey cluster $c(i)$, the gestational window is $W_i = [b_i - 266, b_i]$ under an assumed 266-day gestation. For a radius r , the set of records a child could have been exposed to is

$$\mathcal{E}_i(r) = \{\ell \in \mathcal{L} : \text{dist}(c(i), \ell) \leq r, \delta_\ell \in W_i\}, \quad (2)$$

the intersection of geographic proximity and gestational timing. Exposure is indexed by child rather than by cluster, since two children in the same cluster born months apart have different windows W_i and can differ in exposure holding cluster and radius fixed.

4.1.1 Form

The first condition is that the record describe a locust at a density capable of stripping a field. This uses the phase and presence-type classification already coded in the monitoring data through equation (1) and requires no additional assumption. Records excluded here are not discarded. They are carried through the identical construction below as $\text{Form}(\ell) = \text{benign}$ and used as a falsification group.

4.1.2 Crop stage

The invasion struck at different times across the agricultural calendar, and its timing varied from place to place. The May and June 2020 peak fell while the long-rains maize crop stood under its harvest bar in the western and central highlands, while the short-rains crop stood under its own harvest bar in parts of the east, and over bare ground under land preparation in counties between seasons. The February 2021 peak fell after the short-rains harvest bar had closed in most of the country. Because the calendar differs across Kenya by several months, an arrival date on its own carries no information about whether a crop was lost. Two children conceived in the same month, in clusters 200 km apart, can face swarms on the same dates and face completely different agronomic consequences.

Thus, the second condition is that the arrival find a crop worth destroying. Let $\mathcal{S}(c, \delta)$ be the set of crop stages active in cluster c 's crop calendar on date δ , possibly more than one under staggered planting or

multiple crop groups, and possibly empty. Each stage carries a susceptibility rank,

$$\text{rank}(s) = \begin{cases} 3 & s \in \{\text{vegetative/reproductive, ripening/harvest}\} \\ 2 & s = \text{planting} \\ 1 & s \in \{\text{end-of-harvest, out-of-season}\}, \end{cases} \quad (3)$$

and the governing stage on a cluster-date is the highest-ranked active stage,

$$S(c, \delta) = \arg \max_{s \in \mathcal{S}(c, \delta)} \text{rank}(s), \quad S(c, \delta) = \text{out-of-season if } \mathcal{S}(c, \delta) = \emptyset. \quad (4)$$

A stage is susceptible if its rank exceeds 1,

$$\text{Susceptible}(s) = \mathbb{I}[\text{rank}(s) \geq 2]. \quad (5)$$

Vegetative, reproductive, and ripening stages are susceptible at rank 3. Planting is susceptible at rank 2, since a destroyed planting can be resown at the cost of seed and labor rather than losing the season outright. End-of-harvest and out-of-season are not susceptible, since a swarm arriving after grain is stored or before a crop is planted has nothing standing to destroy. A cluster in a livelihood zone with no crop calendar has $\mathcal{S}(c, \delta) = \emptyset$ on every date and is never susceptible. Damaging-form arrivals excluded by this step are the same insects in the same places at an agronomically irrelevant time, and are carried through as a second falsification group.

“latex The transcribed FEWS NET calendar fixes a normative planting date for each zone, but the rains that actually trigger planting arrive earlier or later than that norm in any given year and cluster, so the calendar is shifted cluster by cluster to track the rain rather than left pinned to its transcribed date. Season onset is detected separately for each cluster, year, and season from dekadal (10-day) CHIRPS rainfall, searched within a season-specific window, dekads 4 to 18 (roughly early February through late June) for the long rains and dekads 25 to 36 (September through December) for the short rains. Let ρ_d denote a cluster’s rainfall total in dekad d of that window. Onset is the first dekad d satisfying three conditions simultaneously:

1. a trigger, $\rho_d \geq 25$ mm;
2. a confirmation, that the one or two dekads immediately following together clear $\sum \rho_{d+1}^{d+2} \geq 20$ mm, so a single wet dekad followed by drought does not count;
3. and a dry-spell check, that at most one of those following dekads itself falls below 5 mm, so a trigger immediately undercut by a dry spell does not count either.

If no dekad in the window satisfies all three, onset is not detected for that cluster-year-season.

A cluster’s climatological onset is the median onset dekad across 1981-2018 for that cluster and season. The transcribed calendar boundaries are shifted, holding stage lengths fixed, by the gap between a given year’s detected onset and that climatological median, rounded to the nearest dekad and capped at six dekads (60 days) in either direction so a single anomalous year cannot pull the calendar implausibly far from its transcribed position; a cluster-year-season where onset could not be detected is left unshifted at its climatological position rather than dropped. A season that begins three weeks late accordingly shifts that cluster’s susceptible window three weeks later rather than leaving it pinned to the calendar date. ““

4.1.3 Verified damage

The third condition is that vegetation actually suffer. NDVI is observed in 16-day composites. Let $z_{c,t}$ be cluster c 's NDVI anomaly z-score in composite t , and let t_ℓ index the composite containing event ℓ 's observation date. Define the mean anomaly in the two composites preceding and the two composites following the event,

$$\bar{z}_{\text{pre}}(\ell) = \frac{1}{2} \sum_{k=1}^2 z_{c(\ell), t_\ell - k}, \quad \bar{z}_{\text{post}}(\ell) = \frac{1}{2} \sum_{k=1}^2 z_{c(\ell), t_\ell + k}, \quad (6)$$

a window spanning roughly 16 to 48 days after the observation, depending on where within its own composite the event falls. Composites with a missing value are dropped from the corresponding average rather than treated as zero. The change in anomaly is

$$\Delta z(\ell) = \bar{z}_{\text{post}}(\ell) - \bar{z}_{\text{pre}}(\ell), \quad (7)$$

and the event is classified as a verified decline if

$$\text{Decline}(\ell) = \mathbb{1}[\Delta z(\ell) \leq -\theta], \quad \theta = 0.25, \quad (8)$$

fixed in advance. A locust record only counts as "verified damage" if the cluster's greenness fell by at least 0.25 standard deviations (θ).

4.1.4 The exposure indicator

Treatment requires all three conditions to hold on a single record,

$$T_i^{(r)} = \mathbb{1}[\exists \ell \in \mathcal{E}_i(r) : \text{Form}(\ell) = \text{damaging}, \text{Susceptible}(S(c(i), \delta_\ell)) = 1, \text{Decline}(\ell) = 1]. \quad (9)$$

Requiring the conjunction on one record rather than across records matters. A swarm in March and a vegetation decline in September are not evidence that the swarm caused the decline. Equation (9) admits only arrivals followed by a fall in the greenness of the same cluster within roughly six weeks.

Equation (8) uses a variable closer to the outcome than the other two conditions, so three safeguards apply. The test in equation (7) is a change in anomaly rather than a level, so a cluster that was already dry does not qualify on dryness alone. The window in equation (6) is short enough that a later seasonal decline unrelated to the event is not attributed to it. Equations (6) to (8) are applied identically to benign records and to damaging-form records arriving outside the susceptible window. If $\text{Decline}(\ell)$ were capturing drought rather than locust damage, these placebo groups would satisfy it at a similar rate and predict outcomes similarly. Section 5.3 reports that they do not.

Three nested measures are defined by dropping conditions from equation (9). Proximity, $T_i^{(1,r)} = \mathbb{1}[\exists \ell \in \mathcal{E}_i(r) : P_\ell = 1]$, is gated on nothing beyond presence and is the closest analogue to the exposure variables used in existing locust research. Damaging form adds equation (1). Damaging form and stage adds equation (5). Then, by construction $T_i^{(r)} \leq T_i^{(3,r)} \leq T_i^{(2,r)} \leq T_i^{(1,r)}$ and $T_i^{(r)} \subseteq T_i^{(3,r)} \subseteq T_i^{(2,r)} \subseteq T_i^{(1,r)}$.

The exposure measure selects a population characteristically different than the unverified measures.⁴

Table 1 compares the disputed children to those both measures agree were exposed. Every gap runs the same direction. Disputed children have mothers with 3.5 fewer years of education, live 7.1 km farther

⁴Table 11 reports counts at every radius.

from a major road, sit 0.66 of a wealth quintile lower, are 19 percentage points less likely to own land, and experienced gestational conditions 0.40 standard deviations drier on SPEI. All of these differences are significant well beyond the 0.1 percent level.

Livelihood composition drives the divergence. Among children both measures agree were exposed, 83 percent live in cropping zones, 15 percent in agropastoral zones, and none in pastoral zones. Among disputed children, 35 percent live in cropping zones and 43 percent in pastoral zones. This is not an empirical finding about where swarms went. This result follows from the construction; a pastoral zone has no crop calendar, so no arrival there can satisfy the stage condition, and no child there can be treated. Proximity flags pastoral children at 64 percent at the 30 km radius, higher than the 48 percent rate in cropping zones, because the swarms arrived in the north and northeast first and stayed longest.

Table 1: Characteristics of disputed children, 30 km radius

	Agreed	Disputed	Difference	<i>p</i>
Mother’s education (years)	9.42	5.88	−3.54	< 0.001
Wealth quintile	3.00	2.35	−0.66	< 0.001
Distance to major road (km)	3.75	10.88	7.13	< 0.001
Owens land	0.545	0.359	−0.186	< 0.001
Owens livestock	0.662	0.672	0.010	0.506
Rural	0.640	0.646	0.006	0.715
Gestational SPEI (3 month)	1.004	0.606	−0.398	< 0.001
Gestational NDVI z-score	0.747	0.840	0.092	< 0.001
<i>Livelihood zone shares</i>				
Cropping	0.831	0.352	−0.480	
Pastoral	0.000	0.426	0.426	
Agropastoral	0.146	0.169	0.024	
Riverine	0.020	0.038	0.018	
Fishing	0.003	0.015	0.011	

Agreed denotes children flagged by both the proximity and verified measures ($N = 1,760$). Disputed denotes children flagged by proximity only ($N = 2,310$). *p*-values from two-sample *t*-tests.

Disputed children have a mean weight-for-height of -0.47 , far below the -0.09 of children both measures agree were exposed and the -0.10 of unexposed children. Read naively, proximity exposure would look devastating and verified exposure harmless. What the raw means record is that the areas proximity adds are drier, poorer, and farther from a road, not that a swarm did anything there.⁵

4.2 Health Outcomes

The DHS measures every sampled child’s length or height and weight under a standardized protocol: recumbent length for children under 24 months, standing height above that age, and weight to the nearest 100 grams, up to 59 months inclusive. DHS converts these raw measurements to three z-scores centrally, before the data is released, against the WHO Child Growth Standards, the international reference built from healthy, breastfed children raised without growth constraints ([WHO Multicentre Growth Reference Study Group, 2006](#)). The conversion uses the WHO’s LMS method: at each value of the conditioning variable (age for HAZ and WAZ, height or length for WHZ) the reference population’s distribution is summarized by three sex-specific parameters, a median M , a coefficient of variation S , and a Box-Cox skewness power

⁵Figure 9 shows the counts at every radius.

L , and a child’s raw measurement X is converted to a z-score

$$Z = \begin{cases} \frac{(X/M)^L - 1}{LS} & L \neq 0 \\ \frac{\ln(X/M)}{S} & L = 0, \end{cases} \quad (10)$$

so that $Z = 0$ is exactly the reference median and $Z = -2$ is two standard deviations below it, in the reference distribution’s own units, regardless of the child’s age. I take DHS’s own pre-computed values directly rather than recomputing them, and treat any value outside ± 6 standard deviations as biologically implausible and set it to missing, following the DHS’s own flagging convention.

Weight-for-height (WHZ) compares a child’s weight to the reference for a child of the same sex and the same attained height or length, with age entering only through which reference curve applies, so it nets out the height a child has already reached and asks only whether her current body mass is proportionate to it. This is what makes it responsive on a short horizon: because it carries no growth history of its own, WHZ falls within weeks or months of a nutritional shock and recovers within months of feeding improving, unlike a measure that accumulates deficits over months or years. An exposure confined to gestation and the months immediately after birth is, by construction, a short-horizon shock, and WHZ is the only one of the three scores built to be sensitive to a short-horizon shock without also carrying the accumulated growth history that would dilute it. The clinical label for WHZ below -2 is *wasted*, and a child below -3 is severely wasted.

Height-for-age (HAZ) and weight-for-age (WAZ) are constructed identically, applying the same LMS transformation in equation (10) against the sex-specific WHO reference curves for length-or-height-for-age and weight-for-age respectively, in place of the weight-for-height curve, with the same ± 6 standard deviation plausibility filter. They differ from WHZ, and from each other, only in which anthropometric ratio and which reference curve they use. HAZ compares height to the reference for a child’s age rather than a child’s height, so it is a cumulative record: a child’s height on any given day reflects every month of growth, or growth faltering, since conception, and a single bad month barely moves it. A low HAZ score is the signature of sustained undernutrition, repeated illness, or a poor environment operating over a long horizon, and the clinical label for HAZ below -2 is *stunted*. WAZ compares weight to the reference for a child’s age; because age enters through both the numerator’s growth history and the denominator’s reference curve, WAZ is a composite of the other two, a child can have a low WAZ score because she is short for her age, because she is thin for her height, or both, and the score alone does not distinguish them. The clinical label for WAZ below -2 is *underweight*.

4.3 Estimation

I estimate

$$y_{ict} = \beta T_i^{(r)} + X'_{ic} \gamma + \alpha_c + \delta_t + \theta_{k(c)} \cdot t + \varepsilon_{ict}, \quad (11)$$

where y_{ict} is the weight-for-height z-score of child i in survey cluster c born in year-month t , $T_i^{(r)}$ is the exposure indicator at radius r defined in Section 4.1, α_c is a survey-cluster fixed effect, δ_t is a birth-cohort fixed effect defined at the year-month level, so that May 2020 and May 2021 occupy separate cells rather than being pooled as the same calendar month, and $\theta_{k(c)} \cdot t$ is a county-specific linear trend in that same year-month index.

The identifying variation is within cluster and across birth cohorts. Two children in the same enumeration area, born eight months apart, face different gestational windows. One window overlaps a verified damage

event within r kilometers and the other does not. Equation (11) compares them.

I include survey cluster and birth-cohort (month $\times$ year) fixed effects. The county-specific linear trend allows counties to sit on different trajectories over the period. A county whose child nutrition was deteriorating for reasons unrelated to locusts, and which the swarms also happened to reach late in the window, would otherwise contribute a spurious negative.

The control set follows Conte et al. (2023). Child controls are sex, birth order, first-born status, twin status, and the birth interval on either side. These are the standard determinants of anthropometry that vary within a cluster and within a birth cohort. Birth spacing in particular is correlated with maternal depletion and with household resources at the time of a birth, and it varies across siblings in a way cluster fixed effects cannot absorb.

Family controls are mother’s education, mother’s height, household head age, household head sex, wealth quintile indicators, and the six-month SPEI over the gestational window. Mother’s height enters because it captures the mother’s own early-life nutrition and is a strong predictor of a child’s growth potential that is unrelated to the locust shock. Wealth quintile indicators enter as a factor rather than a linear term because the relationship with anthropometry is not monotone in the data. Gestational SPEI is the most important of these. Rainfall that is favorable enough to support three generations of locust breeding is also favorable for crops and pasture, so drought conditions are correlated with both the treatment and the outcome. Including SPEI over the child’s own gestational window removes the part of the estimate that reflects climatic water balance rather than the insects.⁶

Standard errors are clustered on a quarter-degree spatial grid, which yields 370 cells across the sample. These are coarser than the survey cluster. A swarm that arrives at one enumeration area almost always arrives at its neighbors, and the downstream effects on vegetation, local food availability, and prices propagate over a similar area. Residuals are therefore correlated across nearby clusters and not only within them, and clustering at the DHS cluster would overstate precision. The quarter-degree grid gives a wider-than-cluster but still local unit. Clustering at the county instead would leave 47 groups, too few for reliable asymptotic reliability and too coarse to absorb genuine variation.

All regressions are weighted by the DHS individual sampling weight.⁷

5 Results

5.1 Health Outcomes and Radii

Table 2 reports the wasting estimates. Columns (1) to (4) use the weight-for-height z-score. Column (1) includes only the fixed effects, column (2) adds the child controls, column (3) adds the family controls, and column (4) adds county-specific linear time trends. Columns (5) and (6) repeat columns (3) and (4) with an indicator for wasting, a weight-for-height score below -2 , as the outcome.

In utero exposure to a verified damage event within 30 km lowers weight-for-height by 0.205 standard deviations in the most conservative specification. This corresponded to a 3.6 percentage point rise in the probability of wasting in column (6), a 51 percent increase over the baseline of 7.2 percent among unexposed children. Without the county trend the estimate is smaller on both measures, 0.185 standard deviations

⁶Occupation and food consumption are excluded from the control set. These variables are unfortunately noisy self-reports in the DHS and are largely missing or the timing requested is irrelevant to the study; one week in most cases.

⁷The weight distribution is slightly right-skewed, see B for more details. Section 5.3 reports the estimate under winsorized weights for robustness.

on weight-for-height and 1.9 percentage points on wasting, a relative increase of 26 percent. The estimate moves by less than 0.02 standard deviations as the child and family controls enter. The standard error is stable near 0.055 throughout and the estimate is significant at the 1 percent level in every column.

Table 2: In utero locust exposure and child wasting, 30 km radius

	Weight-for-height z-score				Wasted	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.192*** (0.056)	−0.196*** (0.055)	−0.185*** (0.055)	−0.205*** (0.057)	0.019** (0.008)	0.036*** (0.010)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,443	7,443	7,416	7,416	7,416	7,416
R^2	0.344	0.351	0.352	0.362	0.261	0.270

All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and birth intervals on either side. Family controls are mother’s education, mother’s height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. Control coefficients are estimated but not reported. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a weight-for-height z-score below −2, with a baseline prevalence of 7.2 percent among unexposed children. Treated children number 1,760.

Among children verified-exposed at the main radius, the number of confirmed damage events during gestation predicts weight-for-height beyond a simple binary indicator of exposure. Table 3 reports the relationship under three functional forms: the raw count of verified damage events entering linearly, the log of one plus that count, and the count capped at 20 to limit the influence of the small number of children exposed to an unusually large number of events. Each additional verified event is associated with a 0.013 standard deviation reduction in weight-for-height, significant at the 5 percent level; the log and capped forms return larger, more precisely estimated reductions of 0.238 and 0.029 standard deviations, both significant at the 1 percent level. All three specifications share the control set and fixed effects of column (3) of Table 2 and additionally control for the number of RAMSES ground visits recorded within 30 km during gestation, which separates the effect of confirmed damage from the effect of monitoring intensity in the same area during the same period.

Table 17 and Table 19 in Appendix A report the same three functional forms for height-for-age and weight-for-age, using the identical verified-exposed sample, control set, and ground-visit control. Weight-for-age moves in the same direction as weight-for-height: an additional verified damage event is associated with a 0.016 standard deviation reduction, significant at the 5 percent level, and the log and capped forms return reductions of 0.280 and 0.037 standard deviations, both significant at the 1 percent level. Height-for-age does not move under any of the three functional forms; the point estimates are negative but none is statistically distinguishable from zero, consistent with the null result already reported for the binary exposure measure.

I estimate the same specifications on the other two standard z-scores and their clinical thresholds. Height-for-age does not move. The point estimate is −0.059 with a standard error of 0.071, and the probability of stunting is unchanged. Weight-for-age falls 0.167 standard deviations, though the probability of being underweight does not change (See Table 15).

Weight-for-height records a deficit of body mass relative to attained height and responds to a nutritional

Table 3: Dose response among verified-exposed children: number of damage events and weight-for-height

Dependent variable: Weight-for-height z-score	Linear	Log	Capped at 20
Verified damage events	−0.013** (0.006)		
Log(1 + verified damage events)		−0.238*** (0.079)	
Verified damage events, capped at 20			−0.029*** (0.010)
Observations	1,459	1,459	1,459

Standard errors clustered on a quarter degree spatial grid in parentheses. ***, ** denote significance at 1 and 5 percent. Sample restricted to children verified-exposed at the main 30 km radius. All columns include the control set and cluster by birth-cohort fixed effects of column (3) of Table 2. I control for the number of RAMSES ground visits during gestation, this counts every recorded visit within 30 km, of any outcome, during the child’s gestation window, and is included to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

shock within weeks, which is why it is the standard indicator of acute malnutrition ([WHO Multicentre Growth Reference Study Group, 2006](#)). Height-for-age records cumulative linear growth from conception onward and responds slowly, over quarters and years. The median child in this sample was 14 months old at measurement and the mean 14.7 months, and the invasion window closed in April 2021. A shock concentrated in gestation and the months around birth has had very little calendar time to enter a height-for-age score, and it has had ample time to enter a weight-for-height score. [Stein and Susser \(1975\)](#) find the same asymmetry in the Dutch famine, where birth weight fell 9 percent while length at birth fell 2.5 percent.

Weight-for-age sits between the two by construction. It is a composite of body mass and linear growth against age, so an acute weight deficit registers there but is diluted by the height component that has not moved. The point estimate on weight-for-age is 90 percent of the weight-for-height estimate, and it does not cross the underweight threshold, which is consistent with dilution rather than with a second distinct effect.

Table 16 and Table 18 in Appendix A report the full stepped-control specification for height-for-age and weight-for-age in the same format as Table 2.

I also estimate the three health outcomes at all nine radii. Figure 3 plots the decay with confidence bands. Weight-for-height is significant at every one. The estimate is largest within 10 km, at 0.312 standard deviations, and falls monotonically in the radius to 0.118 at 50 km. The number of treated children rises from 728 to 2,285 over the same range. This is the pattern a spatially contained shock produces. Widening the buffer adds children who were farther from any verified damage event and dilutes the average effect, and the dilution is smooth rather than abrupt. A result driven by a chance rejection at one radius would not decay this way, and a result driven by a confound correlated with broad geography would not be strongest at the tightest buffer.

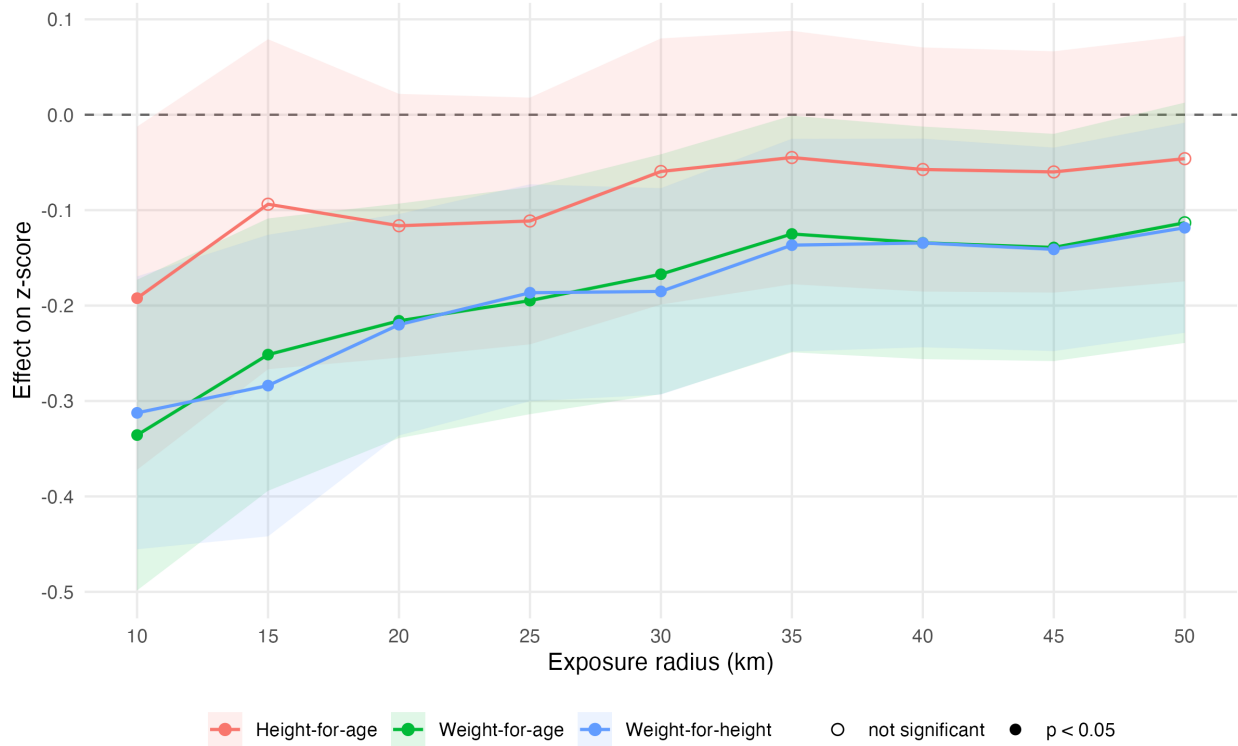


Figure 3: Estimate against radius, by outcome, with 95 percent confidence bands.

5.2 The Exposure Conditions

Figure 4 shows the three conditions of Section 4.1 each regressed at the main radius. Proximity alone is weak. Its weight-for-height estimate is -0.002 with a standard error of 0.050 , and it flags 4,070 children. Restricting to damaging form removes 79 children and moves the estimate to -0.033 . Adding the crop-stage condition removes 1,263 more and moves it to -0.062 , still insignificant. Adding the verification condition removes a further 968 and moves the estimate to -0.185 , significant at the 1 percent level. Two of the three conditions are cheap, using only classifications already present in the monitoring record and a published crop calendar, and neither is sufficient. The signal appears when the measure requires evidence that damage occurred to the vegetation.

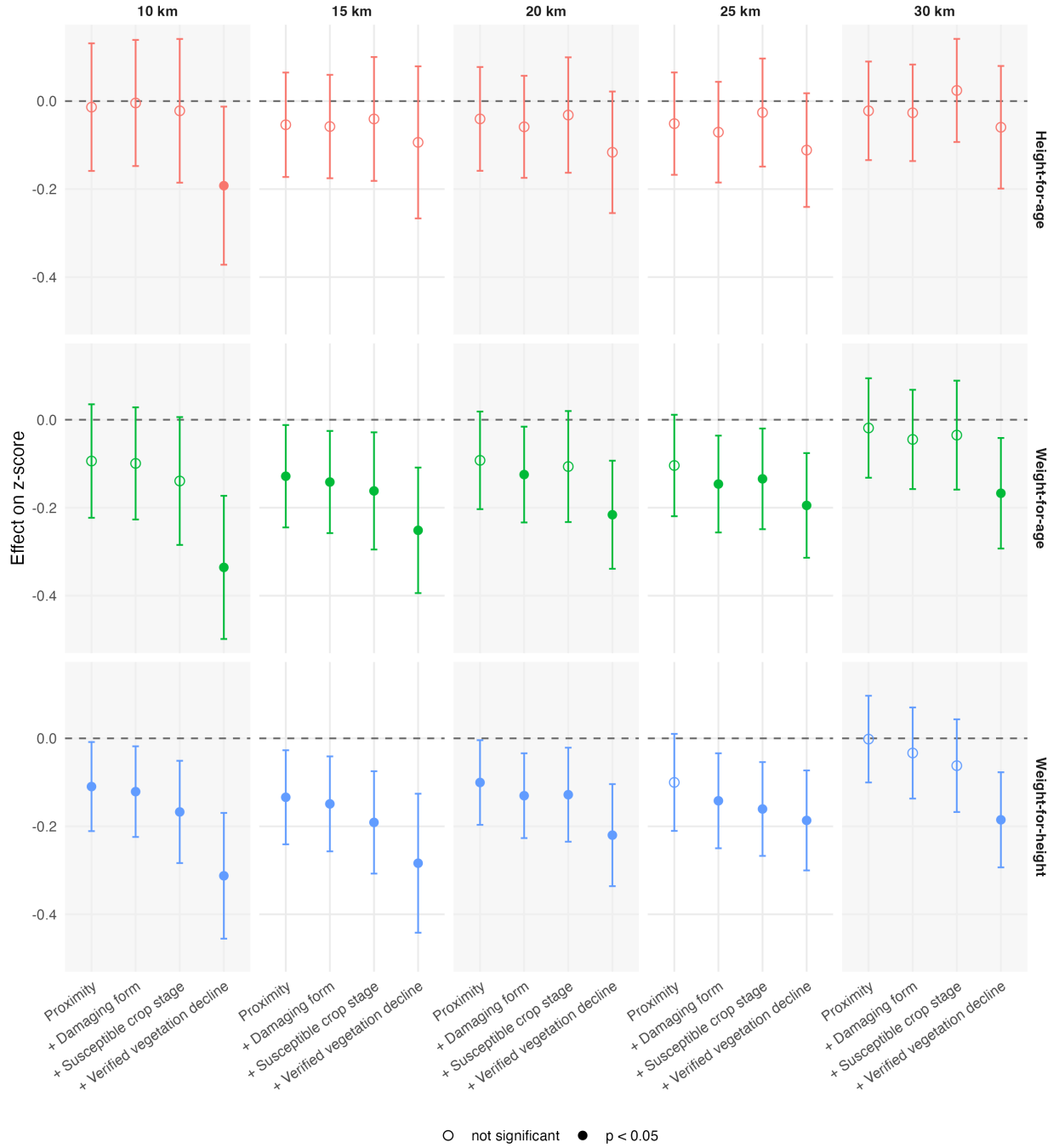


Figure 4: Standard errors clustered on a quarter-degree spatial grid in parentheses. *** denotes significance at 1 percent. Each cell is a separate regression sharing the control set and fixed effects of column (3) of Table 2. The measures nest, so every child flagged by a lower row is also flagged by every row above it.

5.3 Robustness

5.3.1 Falsification Tests

I provide two falsification tests for the exposure conditions. Both run the identical construction on records that cannot have caused crop damage.

The first replaces damaging-form records with solitarious and transiens records and applies the stage and NDVI conditions unchanged. These are locusts at densities incapable of stripping a field. The second uses damaging-form records that arrive outside the susceptible window, the same insects in the same places when there is nothing standing to destroy.

A diagnostic step is required before either test means anything. Applied naively, the solitarious placebo flags children who are also flagged by the real treatment, because a cluster visited by solitarious locusts during a gestational window can be visited by swarms in the same window. At 30 km, 86 percent of the children the naive placebo flags are also treated. A placebo contaminated at that rate will reproduce the treatment effect by construction and tells us nothing. Both placebo groups are therefore purged of any overlap with the treatment before estimation.

Table 7 reports the purged tests. Neither recovers a negative effect on weight-for-height at any radius. The solitarious placebo estimate at 30 km is 0.035 with a standard error of 0.099. The out-of-season placebo estimate is 0.086 with a standard error of 0.065. Across 27 purged placebo cells on weight-for-height, none is negative and significant. If the NDVI condition were simply selecting clusters that happened to dry out, it would select them at a similar rate in these groups and predict the same outcome. It does not.

Permutation inference gives the same answer without relying on the asymptotic variance estimate. Permuting the treatment indicator within survey clusters, 1,000 times, places the 10 km estimate on weight-for-height outside the entire simulated null with a randomization p-value of 0.000. The simulated standard deviation of the permuted estimate is 0.080, close to the clustered analytic standard error of 0.073. Running the same procedure on the solitarious placebo returns a randomization p-value of 0.489.

5.3.2 Drought

A natural threat to identification is a shock to vegetation that coincides with the invasion rather than following from it. The gestational SPEI control absorbs a smooth, cluster-specific measure of the climatic water balance, but it may not absorb all of it, and if verified exposure is systematically correlated with drought conditions that six-month index leaves unmeasured, the wasting estimate could partly reflect a drought effect misattributed to locusts.

Unconditionally, exposed children come from locations with a somewhat higher six-month SPEI on average than unexposed children, 1.27 versus 0.997 (Table 9 in Appendix A), which simply reflects where verified locust damage happened to occur rather than any climate difference caused by exposure. Since the main specification compares children within the same cluster and birth cohort, the gap reverses: exposed children experienced modestly drier gestational windows than their unexposed neighbors in the same cluster and cohort, -0.18 standard deviations on the six-month SPEI index ($p = 0.02$). This within-cluster gap is real, and it runs in the direction that would make a drought confound plausible in principle: a treated group correlated with drier conditions, if drought independently harms weight-for-height, would bias the exposure coefficient toward the negative result this paper reports even absent any locust effect at all.

I remove the SPEI variable and re-estimating at every radius (see Table 8 in Appendix A). If a residual drought channel not fully captured by SPEI were driving the wasting result, dropping it should let that

channel flow unopposed into the treatment coefficient and move it substantially, most plausibly toward zero as the confound is exposed for what it is. It does not. The estimate changes by 0.006 to 0.009 standard deviations at every radius, and in every case moves marginally larger in magnitude rather than smaller. This is the signature of a small, already-absorbed correlation, not of an unmeasured confound: an omitted variable large enough to generate a result of this size would not vanish so completely the moment its only control is removed.

NDVI and SPEI are not interchangeable measures of the same underlying conditions, and the difference is what lets the verification condition speak to locust damage specifically rather than to drought in general. SPEI is a climatic water-balance index, precipitation minus potential evapotranspiration, standardized against a location’s own history; it describes moisture stress acting on every hectare in a cluster roughly uniformly, whether or not a crop or pasture stands on it. NDVI records the greenness of the vegetation actually present on the ground, and it responds to anything that removes standing biomass abruptly, not only to a shortfall in the water available to grow it. A locust swarm defoliating a field over a matter of days produces a sharp, localized drop in NDVI on a timescale a slow-moving drought’s drawdown on the water balance does not resemble; equation (8) is built to catch exactly that signature, a change over roughly six weeks around a specific arrival rather than a seasonal level. This reading rests on one identifying assumption: that no other large-scale consumer of vegetation, a wildlife grazing episode or an unrelated pest outbreak, stripped the same cluster in the same narrow window as a documented locust arrival. Kenya’s rangelands do carry wild herbivores, but a defoliation event that happens to coincide with a verified locust record’s own cluster and six-week window, systematically enough to generate the pattern in Section 5, would have to track the invasion’s own arrival dates and locations rather than occur at random with respect to them.

Thus, another more direct answer to the drought confound is built into the verification condition itself rather than into the control set. Equation (8) tests a change in NDVI anomaly over roughly six weeks around a specific locust arrival, not a level of dryness over the season, so a cluster that is dry for the whole gestational window does not qualify on that dryness alone. The falsification tests above speak to this directly: both placebo groups sit in the same drought-affected places and the same drought years as the real treatment, so if the decline condition were capturing an ambient dry spell rather than an event-specific loss of vegetation, they should satisfy it and predict the outcome at a similar rate. This argument does not depend on which direction the seasonal SPEI comparison runs, since it holds the climatic background fixed and asks only whether a specific arrival on top of it left a specific mark. ⁸

5.3.3 Mortality

The main results conditions on survival. A child who died before the survey has no anthropometric measurement, so the sample-construction filter that requires a valid z-score removes them. If the shock that depresses weight among survivors also raised mortality risk among the most severely affected, the estimates would understate its severity, and if it raised mortality selectively among the children who would have been measured as wasted, the estimates would be biased toward zero.

Additionally, the weaker evidence on an effect on stunting in my analysis might be an artifact of mortality. Stunting is a long-run outcome while wasting is not. Many children might not have survived to an age where the long-run measure of health is observed.

I rebuild the birth panel without the anthropometry filter and re-run the identical exposure construction against it. The 2022 round contributes 8,213 births under this definition against 7,603 with valid measure-

⁸See Figure 8 in Appendix A for more detail.

ments, and 263 of those births, 3.2 percent, did not survive to interview. Table 14 regresses survival on exposure with the same controls, weights, fixed effects, and clustering. Nothing is significant at any radius. Estimates run from -0.000 to 0.007 against a baseline mortality rate of 3.2 percent, and the smallest p-value across the nine radii is 0.38. The three looser measures return the same null at the main radius.

Mortality selection therefore does not explain the weight or height results, and the weight results do not extend to survival. The shock moved children along the acute-malnutrition margin without moving them across the survival margin. The harm is real and measured at a clinical threshold that predicts later morbidity, and it did not, in this episode, translate into excess infant deaths that the survey can detect.

5.3.4 Pre-existing Conditions

If the exposed clusters were simply on a worse trajectory, a treatment constructed the same way in a pre-invasion period should reproduce the result. I test this using the 2014 DHS round, conducted six years before the invasion.

I shift the real locust events back 93 months — seven years and nine months, the gap between the KDHS 2014 and KDHS 2022 fieldwork periods rather than a round number of years — landing the invasion’s calendar pattern on the equivalent point in the 2014 cohort’s actual gestation windows, and re-evaluate the stage and NDVI conditions at the shifted dates using rainfall and NDVI from that period. Event locations and 2014 birth timing are unchanged. Whether a 2014 child’s gestational window overlaps the shifted pattern then varies within cluster exactly as real exposure does.

Across four measures, nine radii, and four outcomes — mortality and the three anthropometric measures — 144 cells in total (Table 10 in Appendix A), four are significant at the 5 percent level, or 2.8 percent, in line with what chance alone implies. The four are mortality under the verified measure at the 30, 35, and 50 km radii and weight-for-age under raw proximity at 50 km, with no shared sign, outcome, or measure to unite them and no adjacent-radius clustering of the kind a genuine gradient would leave. Thus, shifting the real invasion’s calendar pattern onto a cohort it could not have touched produces no more spurious exposure-outcome association than chance alone would predict.

5.4 Potential Mechanisms

5.4.1 Anticipation, Crop Failure and Prices

Conte et al. (2023) argue that a locust invasion reaches children through two channels. An anticipatory channel operates during the growing season, when local prices rise on expected crop loss before any loss is realized. A crop failure channel operates afterward, when the harvest actually fails. For Mali they report that children exposed only to the anticipatory channel are harmed about as much as children who also experienced crop failure, and that both channels show up in local crop prices, with an 8.4 percent price increase during the speculative window that fully accounts for the anticipatory health effect once controlled for.

The authors argue that in Mali, the whole country grows crops on one calendar, one planting season, one harvest, once a year. So when locusts hit in the growing season, there’s a long, shared window, months wide, where every affected family already knows the harvest is in trouble but the crop hasn’t actually failed yet. That’s the “anticipatory” period Conte et al. found: bad news arrives well before the bad harvest does.

In Kenya, rain falls twice a year, so there are two growing seasons, and the exact timing differs across 24 different farming zones. The invasion also lasted 16 months across two separate waves, not one clean season.

Because no single date can serve as the point of common knowledge, I construct an area-specific boundary date, set by the expected completion of that area’s crop.

For a child to count as “anticipatory,” they have to be born strictly after the locust hit (so the family actually knew) but strictly before that same crop’s harvest was supposed to finish. That’s not a season-long window like in Mali, it’s whatever time was left on that specific crop’s clock after the locust already damaged it, often just a few weeks. And it turned out zero children in the entire sample were born in that narrow slice, at any distance we checked.

I only count a locust event as confirmed damage if satellite vegetation data shows a drop within about a month afterward, and that test is more likely to catch locusts that hit a crop when it’s already close to harvest, since damage close to harvest shows up fast and clearly. That naturally squeezes the anticipatory window down to almost nothing.

Price data comes from the World Bank Real Time Prices series for Kenya, 233 markets with monthly maize and bean estimates from 2015 to 2022. The test below uses each reporting market’s own price series directly, with no interpolation to a DHS cluster or a county. A market is treated at radius r if a verified event ever fell within r kilometers of it, and its boundary is the harvest completion date of the earliest such event, the same construction used for the health regression above. Calendar time is measured for every market as quarters relative to that boundary rather than as a shared national quarter, since Kenya has no single national harvest date to index on. A market with no verified event within radius is control throughout its own series. Maize and beans are pooled into one regression rather than tested separately, since the boundary comes from whichever crop made the governing event susceptible, not from a staple-specific recomputation.

$$\log(\text{Price}_{m,t}) = \sum_j \theta_j T_m \times Q_j(t) + \mu_m + \delta_{s,t} + \xi_c \cdot t + \varepsilon_{m,t}, \quad (12)$$

where T_m is the market’s own treatment status at radius r , $Q_j(t)$ is the market’s own relative quarter, μ_m is a market fixed effect, $\delta_{s,t}$ is a staple by year month fixed effect, and $\xi_c \cdot t$ is a county specific linear trend. Standard errors are clustered on market.

Figure 5 plots $\hat{\theta}_j$ at the main 30 km radius. Two and three quarters before the boundary, treated markets run 2.7 to 3.4 percent above control markets, both significant at the 5 percent level. The coefficient falls toward zero in the quarter before the boundary and the quarter of the boundary itself. From two quarters after the boundary the sign reverses. Treated markets run 3.2 percent below control markets two quarters after and 4.1 percent below three quarters after, both significant at the 1 percent level.

The pattern before the boundary is consistent with a genuine anticipatory price response. The pattern after it is not. A crop failure should raise the price of the crop that failed, not lower it, and the post boundary coefficients run the wrong way for a supply shock. Locations near a verified event are also more remote on average, and remoteness in this data predicts thinner real transaction coverage in the price series and a systematically lower recorded price, independent of anything the locusts did. The post boundary result most plausibly reflects that same coverage pattern rather than a market response to realized crop loss. The 2020 to 2021 exposure window also overlaps the first eighteen months of the COVID-19 pandemic in Kenya, which brought its own supply disruption and a documented national maize price spike unrelated to any single locust event, a second reason not to read the post boundary coefficient as a locust price effect.

Read against the health result, the price data speak to the anticipatory channel and not to the crop failure channel. The crop failure effect on weight for height does not need a price signal to operate. A household that loses its own harvest loses food and income whether or not the local market price moves.

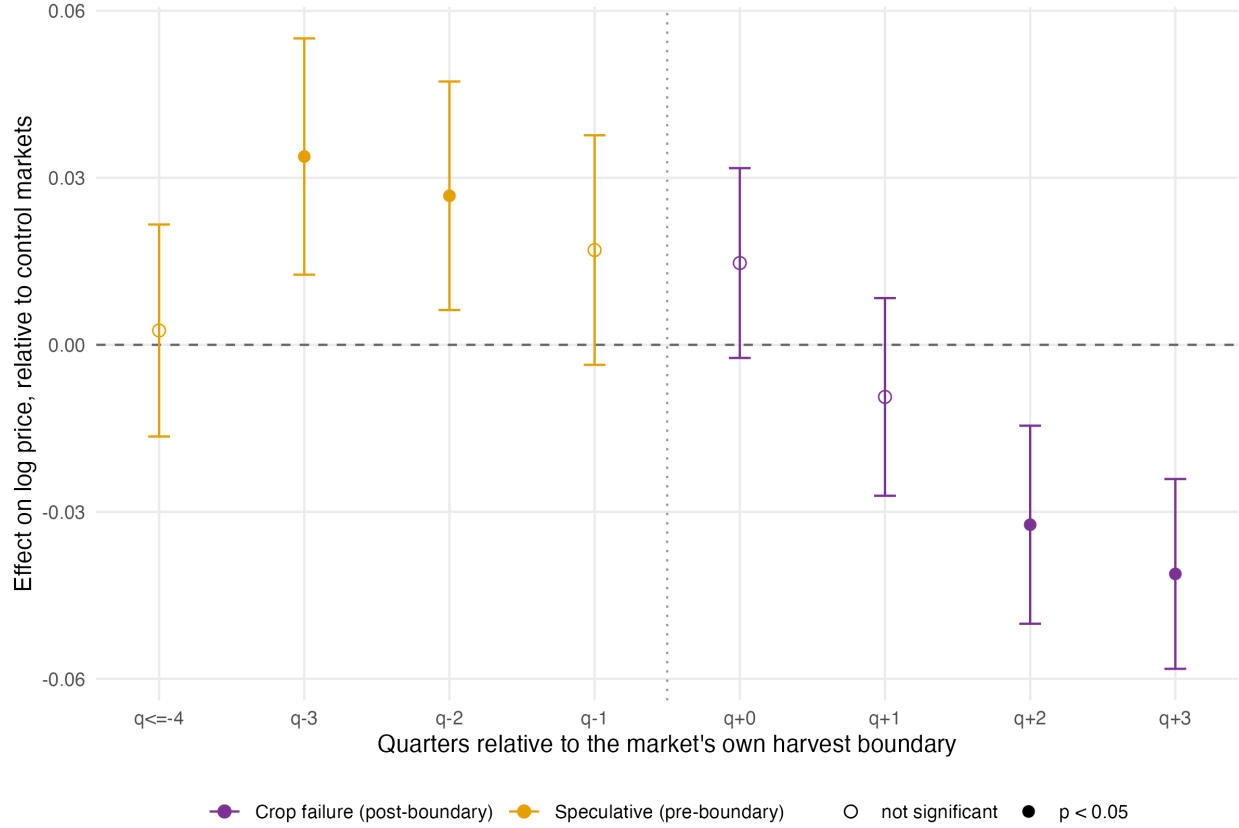


Figure 5: Market level treated versus control price event study. Quarters are relative to each treated market's own harvest boundary. Filled points are significant at the 5 percent level.

5.4.2 Psychological Stress of Proximity

Figure 6 shows estimates at restricted samples - pastoral and fishing zones - and regresses each outcome on proximity exposure at every radius. Nothing is significant anywhere. Point estimates are positive as often as negative and never exceed 0.12 standard deviations, on 642 to 1,203 treated children. These are the households that were closest to the swarms and most often flagged by proximity, and they show no anthropometric response.

This tests two things. First, it conceptually rules out a general channel operating through the presence of locusts on mental health rather than through physical deprivation from hunger. It also shows that the proximity measure's null at the main radius is not a power problem in disguise. Proximity does not detect an effect because in a large share of the children it flags, there was no crop-loss mechanism to detect.

I run a similar test on a sample that was only treated when crops are not susceptible. Gregarious swarms of locusts that arrive outside a susceptible crop stage, have no crop-loss mechanism at all, since nothing was standing in the field to threaten. These are "scare" exposures. Run on the cropping zone sample the health effects are null.⁹

⁹A livelihood zone with no crop calendar can never produce a crop-stage nor verified damage record, so every damaging-form encounter a pastoral child has is biology-only by construction, and pastoral children are already poorer, more remote, and on drier gestational conditions than the rest of the sample.

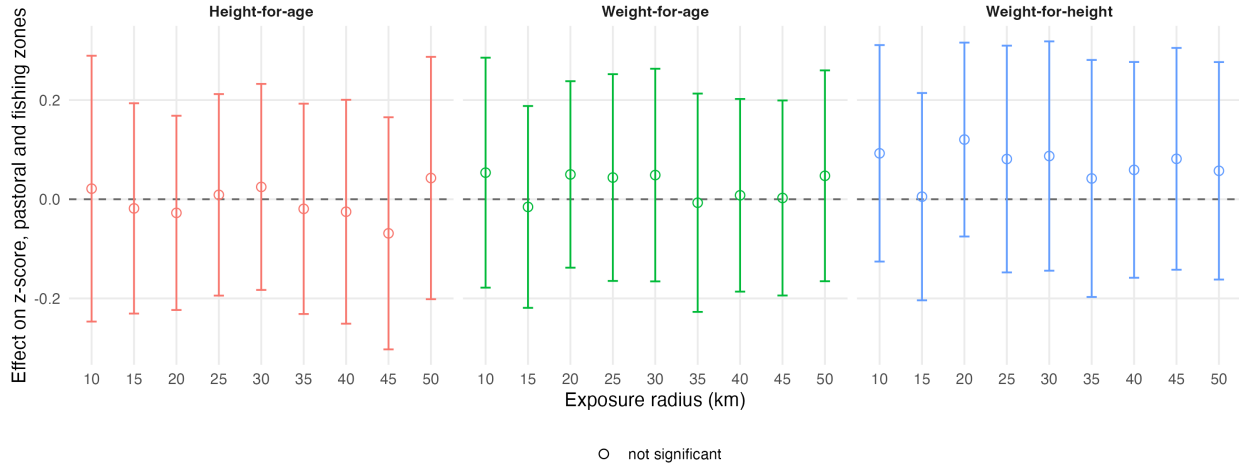


Figure 6: Proximity exposure in pastoral and fishing zones. Standard errors clustered on a quarter-degree spatial grid in parentheses. No estimate is significant at the 10 percent level. Sample restricted to children in pastoral and fishing livelihood zones. Each cell is a separate regression sharing the control set and fixed effects of column (3) of Table 2.

5.5 Heterogeneity

5.5.1 Which Households Are Hit Hardest

Table 4 compares exposed children who are actually wasted to exposed children who are not, at the main 30 km radius. Wasted cases have mothers with 3.0 fewer years of education, sit 0.69 of a wealth quintile lower, and live 1.3 km farther from a major road. Underweight cases show a similar gradient, with mothers 1.2 cm shorter and a 13 percentage point higher rural share. Gestational SPEI is modestly lower among wasted cases (-0.23 standard deviations, $p = 0.021$) and gestational NDVI is modestly higher among underweight cases ($p = 0.015$), but both gaps are small next to the education, wealth, and distance gaps above, which are larger and more consistently significant; the balance of evidence still points to household capacity rather than differential exposure driving who among the exposed ends up a case, though the clean null on climate that held at 15 km does not hold as cleanly here.

Table 4: Exposed children who are wasted versus exposed children who are not, 30 km radius

	Wasted	Not wasted	Difference	p
Mother's education (years)	6.56	9.58	-3.02	< 0.001
Wealth quintile	2.34	3.04	-0.69	< 0.001
Distance to major road (km)	4.99	3.69	1.30	0.073
Rural	0.700	0.637	0.063	0.210
Owns land	0.444	0.551	-0.106	0.052
Mother's height (cm)	160.43	160.63	-0.20	0.591
Gestational SPEI (3 month)	0.787	1.016	-0.228	0.021
Gestational NDVI z-score	0.702	0.750	-0.048	0.571
Age at measurement (months)	14.58	15.73	-1.16	0.079

Sample restricted to children flagged as exposed at 30 km. Wasted denotes a weight-for-height z-score below -2 ($N = 90$). Not wasted denotes the remainder of the exposed group ($N = 1,669$). p -values from two-sample t -tests.

Table 5 reports these interactions formally, for all three anthropometric outcomes rather than weight-

for-height alone, from two separate specifications.

The top panel estimates, separately for each outcome $y \in \{\text{WHZ, HAZ, WAZ}\}$ and each candidate characteristic $Z_i \in \{\text{agricultural occupation, owns land, rural, distance to major road}\}$, twelve regressions of the form

$$y_{ict} = \beta T_i^{(r)} + \lambda Z_i + \eta (T_i^{(r)} \times Z_i) + X'_{ic} \gamma + \alpha_c + \delta_t + \varepsilon_{ict}, \quad (13)$$

one regression per (characteristic, outcome) pair, so the four rows in a single column of the top panel are twelve distinct models, not four coefficients from one shared regression. $T_i^{(r)}$ is exposure at the main radius, and X_{ic} , α_c , and δ_t are the same family-and-child control vector, survey-cluster fixed effect, and birth-cohort fixed effect as column (3) of Table 2; the county-specific trend in equation (11) is not included here. The reported coefficient is η , the interaction between exposure and the characteristic; β and λ are not shown.

The bottom panel instead estimates one regression per outcome that includes every livelihood zone jointly. Let $D_{iz} = \mathbb{1}[\text{zone}_i = z]$ for each zone $z \in \{\text{pastoral, agropastoral, riverine, fishing}\}$, with cropping omitted as the reference zone:

$$y_{ict} = \beta T_i^{(r)} + \sum_z \lambda_z D_{iz} + \sum_z \eta_z (T_i^{(r)} \times D_{iz}) + X'_{ic} \gamma + \alpha_c + \delta_t + \varepsilon_{ict}, \quad (14)$$

with the same control set and fixed effects as equation (13). Cropping's own exposure effect is the coefficient β on $T_i^{(r)}$ alone, not reported in the table; each η_z is that zone's exposure effect *relative to* cropping's, so the three reported rows in a column all come from the same regression and are directly comparable to one another, unlike the top panel. The pastoral interaction is not estimable and is dropped: no pastoral child is ever treated under the verified-damage measure, so $T_i^{(r)} \times D_{i,\text{pastoral}}$ has no variation to identify η_{pastoral} from. Rural residence blunts the exposure effect significantly on weight-for-height and weight-for-age, an offset of 0.24 standard deviations on each, significant at the 1 percent level, though not on height-for-age; agricultural occupation moves the same direction on weight-for-height alone, at the 10 percent level. Land ownership and distance to a road show no significant interaction with exposure on any outcome. These four characteristics are highly correlated with one another in this sample – rural residence alone correlates -0.56 with wealth quintile and -0.28 with livestock ownership – so they plausibly proxy a single underlying divide between subsistence-oriented and market-dependent households rather than four independent channels. Entering all four jointly in one regression, rather than one at a time (Table 12 in Appendix A), supports that reading: only rural residence survives, and only at the 10 percent level on weight-for-height and weight-for-age, while agricultural occupation, land ownership, and distance to a road all lose significance once rurality is controlled for directly. Being rural blunts the exposure effect on short-run weight measures, but the data cannot distinguish which specific asset, land, livestock, occupation, or market access, carries that protection.

The bottom panel of Table 5 interacts exposure with livelihood zone, with cropping zones as the reference category rather than each zone's own isolated treatment slope, since cropping holds 1,463 of the 1,760 treated children at the main radius and is the natural baseline for the comparison. Riverine zones take a significant additional hit beyond the cropping baseline on all three outcomes: an extra -0.25 standard deviations on weight-for-height ($p = 0.057$), -0.39 on height-for-age ($p = 0.041$), and -0.37 on weight-for-age ($p = 0.005$).

This is not a restatement of the rural finding above: riverine households are, if anything, less rural than the rest of the exposed sample, 31 percent against 61 to 77 percent in every other zone, and of middling wealth between the pastoral and cropping extremes, so the riverine gap survives holding wealth quintile fixed. It instead reflects riverine livelihoods' specific dependence on flood-recession agriculture. Riverine households in the underlying livelihood-zone data source roughly 55 percent of their food from their own

production, against about 45 percent in cropping zones, not because they grow fewer crops than a cropping-zone household (riverine zones report four to five own-produced staples, comparable to or above the cropping-zone average), but because they have far less of a market cushion when a harvest is lost: a cropping-zone household already buying close to half its food can substitute purchases for a damaged crop more easily than a riverine household that is not used to buying its way out of a bad harvest. The riverine estimate is stable across the radius range the paper’s main results are built on: re-estimated at every radius from 10 to 50 km (Table 13 in Appendix A), the coefficient is negative and mostly significant from 15 to 35 km and collapses to a small, sign-flipped, statistically indistinguishable-from-zero estimate at 40 km and beyond – consistent with a local damage effect that fades with distance rather than a spurious pattern.

Agropastoral zones show a milder but statistically significant protective effect relative to cropping on all three outcomes, resting on 256 treated children across 80 clusters and 11 counties, though concentrated in the Western Agropastoral zone. Fishing zones return a large, significant positive coefficient on all three outcomes, but it rests on six treated children in a single grid cell in one fishing zone on Lake Victoria and should not be read as evidence of anything beyond that one location.

Table 5: Heterogeneity: 30km exposure interacted with baseline characteristics and livelihood zone

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score
<i>Baseline characteristic</i>			
Agricultural occupation	0.212* (0.113)	0.046 (0.134)	0.167 (0.127)
Owens land	0.101 (0.080)	0.008 (0.083)	0.074 (0.080)
Rural	0.239*** (0.087)	0.111 (0.114)	0.238*** (0.088)
Distance to major road (km)	0.011 (0.010)	0.018 (0.011)	0.018* (0.010)
<i>Livelihood zone (relative to cropping)</i>			
Agropastoral	0.250* (0.136)	0.325** (0.146)	0.333*** (0.117)
Riverine	-0.249* (0.130)	-0.390** (0.191)	-0.372*** (0.132)
Fishing	0.184* (0.096)	0.574*** (0.096)	0.379*** (0.109)

Panel A: each cell reports the coefficient on exposure (30km) interacted with the row characteristic from a separate regression of the column outcome on exposure, the row characteristic, their interaction, and the control set and fixed effects of column (3) of Table 2. Panel B: each cell reports exposure interacted with livelihood zone from a single regression per outcome, cropping as the reference level, with the same control set and fixed effects; each coefficient is that zone’s exposure effect relative to cropping, not an own isolated slope. Pastoral zones are excluded because no pastoral child is ever treated under the verified-damage measure. At the main radius, 1,463 of the 1,760 treated children are in cropping zones, 256 in agropastoral, 35 in riverine, and 6 in fishing; the riverine and fishing coefficients rest on far fewer treated children than the cropping and agropastoral estimates and should be read with that in mind, though Table 13 shows the riverine gap is stable across radius. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

5.5.2 Timing Within Gestation

Figure 7 splits verified exposure by the trimester in which the qualifying event falls, at every radius rather than only the two originally reported, on weight-for-height and weight-for-age.

First-trimester exposure is never significant on either outcome, at any radius. Third-trimester exposure is largest and most significant from 10 to 25 km, but its significance fades steadily with radius and disappears entirely past 30 km on both outcomes. Second-trimester exposure does the opposite: it strengthens as the tighter-radius third-trimester estimates fade, and remains significant at the 1 percent level on weight-for-height from 20 km all the way to 50 km. At the paper’s own main radius, 30 km, second-trimester exposure is now the larger and more significant of the two, -0.300 ($p < 0.01$) against third trimester’s -0.176 ($p < 0.05$) on weight-for-height, and -0.260 against -0.213 on weight-for-age, both significant at the 1 percent level. The pooled estimate in Table 6 is therefore not a third-trimester result diluted by pooling, as the narrower table suggested; it is closer to a second-and-third-trimester result, with the balance between the two shifting toward the second trimester as the radius widens.

Height-for-age offers one further nuance. The pooled estimate in Table 6 shows no significant effect on height-for-age at the main radius, which Section 5 reads as the signature of an acute rather than chronic shock. Split by trimester, this is not the whole story: third-trimester exposure is significant on height-for-age from 10 to 30 km (-0.281 to -0.163 , all at the 5 percent level or better), while first- and second-trimester exposure are never significant at any radius. The pooled null is therefore an average of a real, radius-limited third-trimester effect and two null trimesters, not a uniform absence of any linear-growth response. This is not a contradiction of the acute-shock reading. The third trimester is also the window of most rapid fetal linear growth. [Stein and Susser \(1975\)](#) find a smaller but real effect of third-trimester famine exposure on length at birth alongside the larger effect on birth weight; a shock concentrated in the trimester when skeletal growth is fastest is the one trimester in which even an acute insult has a plausible route to affecting height, and it is exactly the one that does.

A caloric-deprivation shock predicts loading on the second and third trimesters rather than the first: birth weight and its correlates are gained overwhelmingly in the second half of gestation, so a shock to household food availability should be least visible when it falls early and most visible when it falls late ([Currie, 2011](#)). This is a different signature from shocks that work through other channels: [Almond et al. \(2009\)](#) find that prenatal Chernobyl fallout’s cognitive costs concentrate at the gestational weeks tied to neural development, and [Almond and Mazumder \(2011\)](#) find that Ramadan fasting, a caloric but circadian rather than nutritional shock, loads on the first trimester instead. The trimester a shock loads on is diagnostic of its channel, not a fixed biological fact, and the pattern here is the one a maternal energy-balance channel should produce. Currie’s own caution matters as well: because fetal weight gain concentrates in the third trimester, weight-for-height can behave as an insensitive marker of whatever trimester is developmentally critical, understating first-trimester damage that manifests elsewhere. Figure 7’s null on first-trimester exposure is therefore evidence against a first-trimester effect on weight-for-height specifically, not evidence that earlier exposure has no consequences for this cohort.

The specific balance my paper finds; that third trimester dominant at tight radii and second trimester dominant and more persistent as the radius widens, reflects the shock’s own design rather than gestational biology alone. [Stein and Susser \(1975\)](#) isolate a clean third-trimester effect because the Dutch famine was a single, sharply dated shock lasting a few months, so every child’s trimester of exposure maps onto the same calendar window. The 2020–2021 invasion described in Section 2 was not that kind of shock: it ran sixteen months across two waves and five growing seasons, so “third-trimester exposure” does not pick out one calendar period the way it does in a single-harvest famine, and widening the radius pulls in children from a more heterogeneous set of clusters and arrival dates with correspondingly less reason to share a trimester. [Abiona and Ajefu \(2023\)](#) and [Currie and Rossin-Slater \(2013\)](#) document the same instability in Sierra Leone

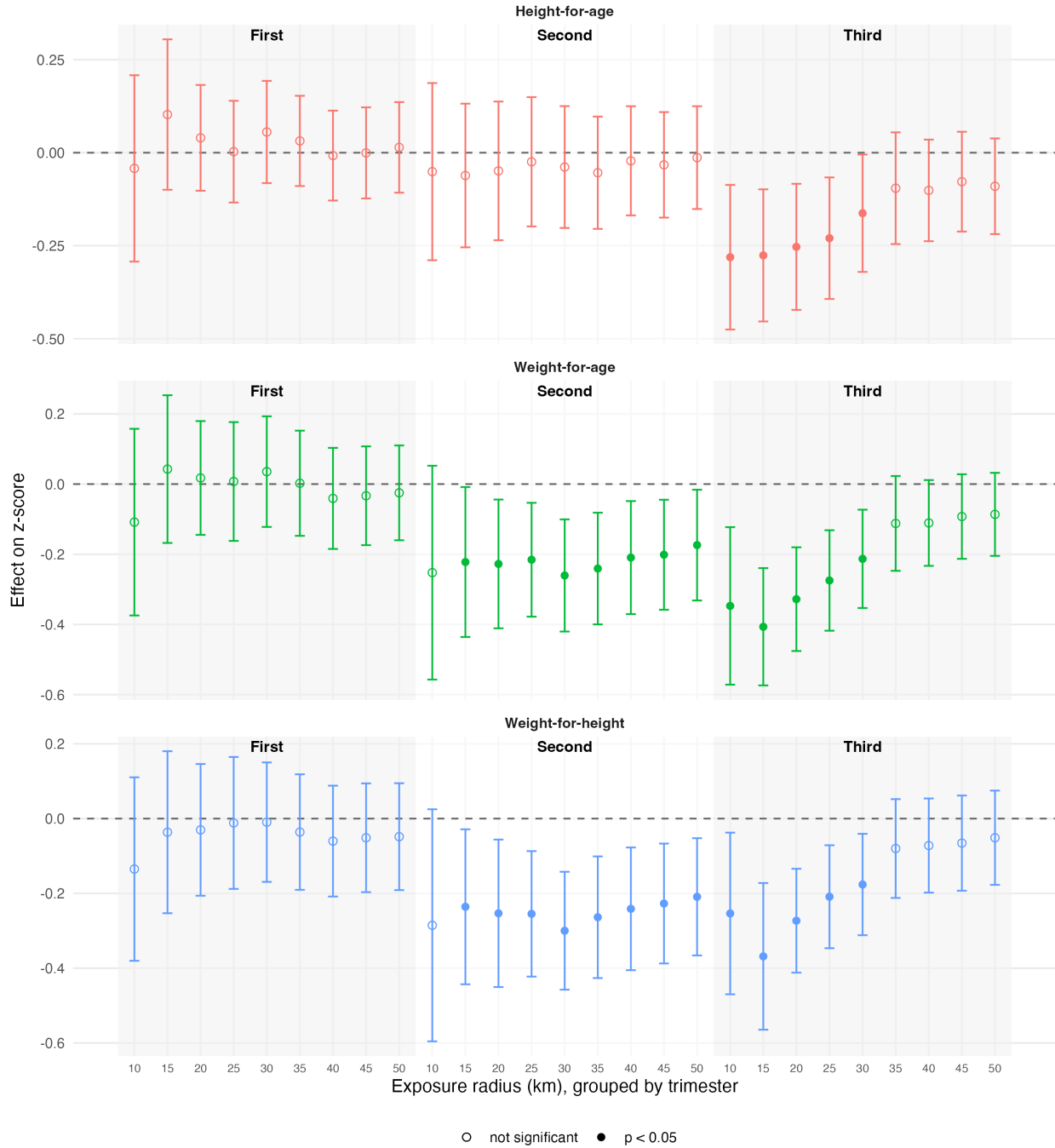


Figure 7: Standard errors are clustered on a quarter-degree spatial grid. ***, **, * denote significance at 1, 5 and 10 percent. Each cell is a separate regression on an indicator for verified exposure occurring during that trimester, sharing the control set and fixed effects of column (3) of Table 2.

and after a US hurricane, respectively: the gestational window a climate-linked shock loads on shifts with the measure and the sample in a way the single-event famine literature does not prepare you for.

The closest direct comparison, [Bauer and Mburu \(2017\)](#) on drought in Marsabit. This the same pastoralist-adjacent region this paper's disputed sample draws from. They find second- and third-trimester

exposure damaging child health, not first, as measured by mid-upper-arm circumference. [Hoddinott and Kinsey \(2001b\)](#)’s Zimbabwe evidence on drought-driven growth faltering similarly implicates the back half of gestation. Taken together, the second-and-third-trimester result looks like a recurring feature of agricultural income shocks in low-income farming and herding settings. The same logic closes the loop on height-for-age: [Stein and Susser \(1975\)](#) find a small but real reduction in length at birth from third-trimester famine exposure alongside the much larger birth-weight effect, because the third trimester is also the window of fastest fetal linear growth, the one trimester in which even a short shock has a plausible route to a skeletal outcome.

6 Discussion

Kenya’s 2020–2021 locust invasion harmed the children it reached. Within 30 km of a verified damage event, in utero exposure lowered weight-for-height by 0.19 to 0.21 standard deviations and raised the probability of wasting by 1.9 to 3.6 percentage points against a baseline of 7.2 percent. The estimate holds at every radius from 10 to 50 km, grows as the buffer tightens, loads on the third trimester, and concentrates among the poorest and most remote households within the exposed group. Height-for-age does not move, which is what an acute shock should do to children whose median age at measurement is 14 months.

The result is visible because of how exposure is defined. A measure that records locust presence within 30 km of a household recovers a coefficient of -0.002 on the same children, the same outcome, the same controls, and the same fixed effects. Requiring the record to describe a gregarious form and to arrive in a susceptible crop stage moves the estimate to -0.062 , still indistinguishable from zero. Requiring a measured decline in vegetation over the following six weeks moves it to -0.185 and to significance at the 1 percent level. Two thirds of the children a proximity measure flags at 30 km show no evidence that anything happened to the vegetation around them, and 43 percent of those live in pastoral zones where no crop calendar applies. Proximity to a swarm does not predict harm because a swarm that arrives over bare ground, or over a rangeland, is not the same event as a swarm that arrives over a ripening field.

The paper’s main limitation concerns the price channel. Section 5.4.1 shows that essentially no children in this sample were born in the narrow anticipatory window [Conte et al. \(2023\)](#) identify in Mali, and that the market-level price event study’s post-boundary movement is more plausibly a market-coverage artifact, remote treated markets reporting fewer real transactions, exposed to trade with other countries and COVID-era price disruptions, than a genuine crop-failure supply response. This paper cannot therefore cleanly separate a price-mediated channel, either anticipatory or realized, from the direct loss of a household’s own harvest. The health estimates capture whatever combination of the two actually operated; they do not decompose it the way the price data would ideally allow.

The results have implications for policy as well as for future research. A relief program that defines exposure by proximity enrolls 4,070 children where a damage-verified rule enrolls 1,760, and the 2,310 it adds are drier, poorer, more remote, hold less land, and are disproportionately pastoral. That is not a wider net over the same population; it is a different net entirely, and the two rules would end up funding different districts. Whether the wider net is the right choice depends on what the program is actually trying to do. A household that lost a specific harvest is well served by aid sized to replace that harvest. A pastoral household with no crop to lose in the first place, but that is nonetheless poor and remote, needs something closer to ordinary income support rather than compensation for a loss it never had. A targeting rule built only on “was a swarm nearby” cannot tell these two situations apart, and in practice it will end up directing harvest-replacement aid toward households that had no harvest to replace, while glossing over the different

and more persistent kind of need pastoral communities actually face.

For research, a measure built only from where the hazard was recorded ignores that biology entirely, and it is not a coincidence that it also finds nothing. The broader lesson for economists studying natural disasters is to let the natural science of the hazard, discipline how we assess their consequences. A flood, a drought, a pest outbreak, or a storm behaves according to its own physical or biological logic, and a measure that ignores that logic is measuring where the hazard was observed, not what it did.

Finally, in the context of this particular research question, two extensions would sharpen this picture further. In the short run, enumeration with shorter gaps during a natural disaster could well uncover behavioral response that might better inform policy; whether households choose to sell livestock, their dietary and migration responses, and if they kept harvests that they might sold for themselves to cope. In the longer run, following cohorts forward could show whether the deficits measured in this literature persisted into cognitive development, schooling, and eventual earnings of the children who live through these shocks.

A Supplemental Tables and Figures

Table 6: Treatment by Radius and Health Outcome

Radius	Weight-for-height z-score	Weight-for-age z-score	Height-for-age z-score	Wasted	Treated
10 km	−0.312*** (0.073)	−0.336*** (0.083)	−0.192** (0.092)	0.024** (0.010)	728
15 km	−0.284*** (0.081)	−0.251*** (0.073)	−0.094 (0.088)	0.028*** (0.009)	1,067
20 km	−0.220*** (0.059)	−0.216*** (0.063)	−0.116* (0.070)	0.021** (0.008)	1,333
25 km	−0.187*** (0.058)	−0.195*** (0.061)	−0.111* (0.066)	0.021** (0.008)	1,529
30 km	−0.185*** (0.055)	−0.167*** (0.064)	−0.059 (0.071)	0.019** (0.008)	1,760
35 km	−0.137** (0.057)	−0.125** (0.063)	−0.045 (0.068)	0.016* (0.008)	1,911
40 km	−0.134** (0.056)	−0.134** (0.062)	−0.057 (0.065)	0.015* (0.008)	2,046
45 km	−0.141*** (0.054)	−0.139** (0.061)	−0.060 (0.064)	0.016** (0.008)	2,164
50 km	−0.118** (0.056)	−0.113* (0.064)	−0.046 (0.066)	0.013 (0.008)	2,285

Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Each cell is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Sample sizes are 7,416 for weight-for-height and wasting, 7,421 for weight-for-age, and 7,424 for height-for-age.

Table 9 compares exposed and unexposed children on the covariates available before treatment. The unconditional comparison looks like a typical cross-section of Kenyan households sorted by geography: exposed children have better-educated mothers, more land, higher wealth, and live closer to a road, and twelve of the nineteen variables differ at the 5 percent level. Almost none of this survives conditioning on the same cluster and birth-cohort structure the main specification uses. Mother’s education, wealth, land ownership, and household size all become indistinguishable from zero, which is what the cluster fixed effect is for: it removes exactly the geographic sorting that produces the raw gaps. Two covariates remain imbalanced after conditioning, twin status and mother’s height, both plausible chance findings given the number of tests, and neither is a covariate with an obvious channel to weight-for-height beyond what the control set already absorbs. Rural status and distance to a road are fixed within a cluster and are absorbed entirely by the cluster fixed effect, so no adjusted comparison is possible for either. Gestational SPEI and NDVI remain significant after conditioning, in the opposite direction from their raw comparison: exposed children experience modestly drier and less green gestational windows than their unexposed neighbors within the same cluster and cohort. Section 5.3 addresses this directly.

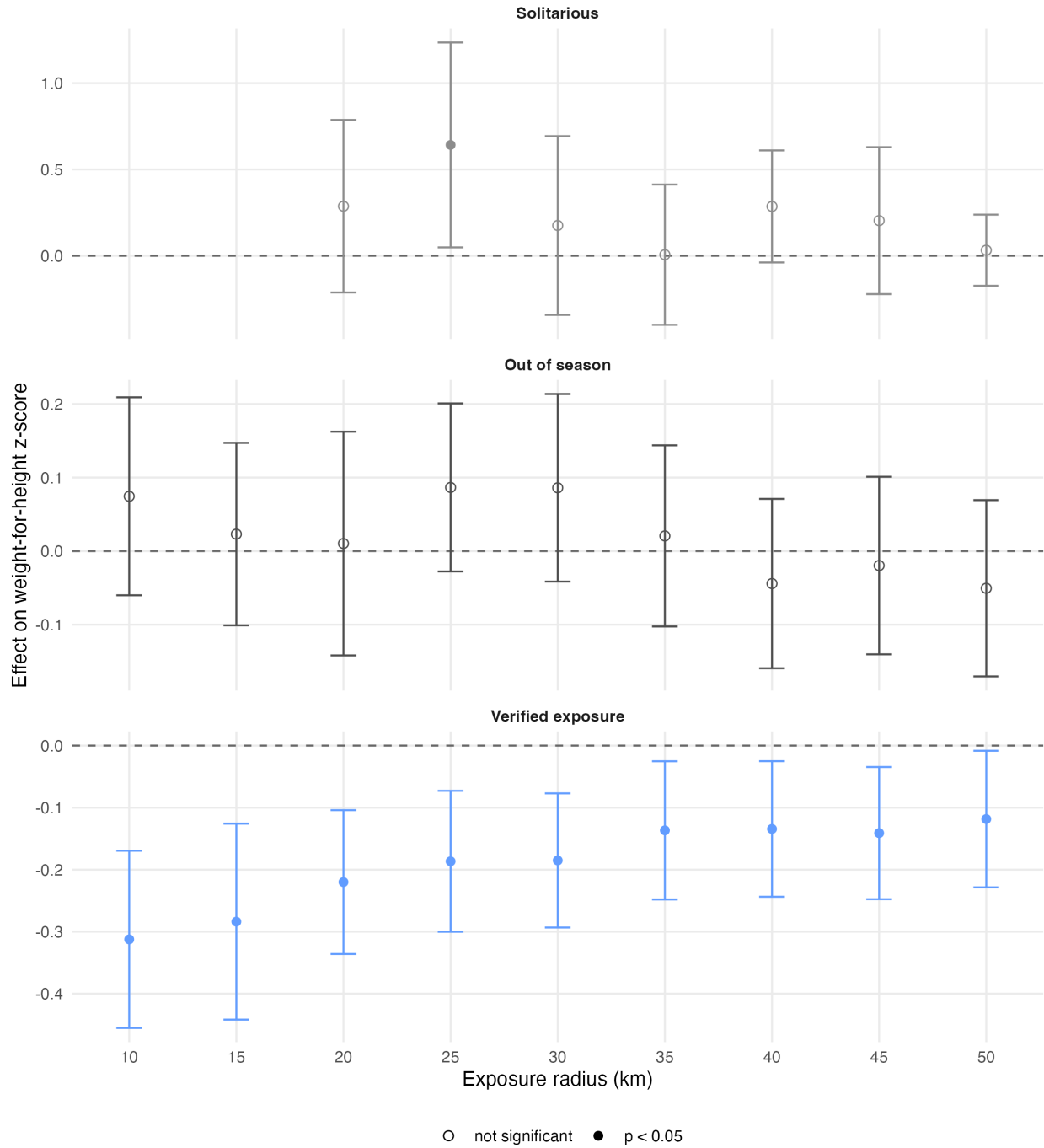


Figure 8: Falsification tests and verified exposure on weight-for-height, by radius. Solitarious and out-of-season estimates are purged of overlap with verified exposure at the same radius, as described in Section 5.3. Filled points are significant at the 5 percent level; hollow points are not. Y-axes are independent across panels and are not comparable to one another in scale. No solitarious estimate is reported at 10 or 15 km: purging leaves 11 and 19 children respectively, both below the 20-observation floor applied throughout this paper, against 56 and 119 before purging (80 and 84 percent of each group is contaminated with genuine verified exposure at that radius). The verified-exposure panel reproduces the D4_verified column of Table 6.

Table 7: Falsification tests: purged placebo estimates on weight-for-height, by radius

Radius	Any-form solitary/transiens	Form + stage + NDVI	Out-of-season damaging form
10 km	0.064 (0.103)	—	0.075 (0.069)
15 km	0.086 (0.094)	—	0.023 (0.063)
20 km	0.122 (0.080)	0.287 (0.255)	0.010 (0.078)
25 km	0.138* (0.078)	0.642** (0.303)	0.087 (0.058)
30 km	0.035 (0.099)	0.176 (0.264)	0.086 (0.065)
35 km	0.026 (0.090)	0.006 (0.207)	0.021 (0.063)
40 km	0.088 (0.079)	0.286* (0.166)	−0.044 (0.059)
45 km	0.078 (0.085)	0.204 (0.217)	−0.020 (0.062)
50 km	0.023 (0.072)	0.032 (0.105)	−0.050 (0.061)

Outcome is weight-for-height (WHZ). Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All three columns are purged of overlap with verified (D4) exposure at the same radius, as described in Section 5.3, and share the control set and fixed effects of column (3) of Table 2. Any-form solitary/transiens is solitary and transiens locust records purged of D4 overlap; Form + stage + NDVI additionally requires the crop-stage and vegetation-decline conditions of equation (9) and is not estimated at 10 or 15 km, where purging leaves fewer than 20 children (see Figure 8); Out-of-season damaging form is damaging-form records purged of D4 overlap that arrive outside the susceptible crop stage. 27 nominal cells, 25 estimated after the 20-observation floor; none is negative and significant.

Table 8: Wasting Estimates with and without the drought control

Radius	With SPEI-6	Without SPEI-6	Treated
10 km	−0.312*** (0.073)	−0.320*** (0.071)	728
15 km	−0.284*** (0.081)	−0.290*** (0.078)	1,067
20 km	−0.220*** (0.059)	−0.226*** (0.058)	1,333
25 km	−0.187*** (0.058)	−0.193*** (0.057)	1,529
30 km	−0.185*** (0.055)	−0.192*** (0.054)	1,760
35 km	−0.137** (0.057)	−0.145*** (0.056)	1,911
40 km	−0.134** (0.056)	−0.143*** (0.054)	2,046
45 km	−0.141*** (0.054)	−0.149*** (0.053)	2,164
50 km	−0.118** (0.056)	−0.128** (0.055)	2,285

Outcome is the wasting z-score (WHZ). Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. *With SPEI-6* repeats the specification of column (3) of Table 2. *Without SPEI-6* is identical except for dropping the six-month gestational SPEI control, the only variable in the specification that measures drought directly. Observations are 7,416 in every cell. The three-month SPEI index is not reported here or used as a control anywhere in this paper, since it is highly collinear with the six-month index at the cluster level by construction and adds no independent information.

Table 9: Covariate balance, 30 km radius

	Control	Treatment	Raw difference	FE-adjusted	(SE)
<i>Child</i>					
Female	0.490	0.476	−0.014	−0.026	(0.024)
Birth order	3.38	2.75	−0.629***	−0.054	(0.096)
First born	0.246	0.311	0.066***	0.019	(0.024)
Twin	0.031	0.021	−0.010**	—	
Birth interval, prior (days)	34.5	36.0	1.56	0.445	(2.03)
Birth interval, next (days)	0.081	−0.462	−0.543***	—	
<i>Mother and household</i>					
Mother’s education (years)	7.13	9.42	2.30***	−0.044	(0.148)
Mother’s height (cm)	161.0	161.0	−0.174	0.360**	(0.183)
Household head age	38.9	38.3	−0.576*	−1.04*	(0.563)
Female household head	0.291	0.268	−0.023*	−0.020	(0.022)
Household size	6.06	5.38	−0.683***	−0.020	(0.129)
Wealth quintile	2.48	3.00	0.519***	0.021	(0.035)
<i>Assets and access</i>					
Owns land	0.462	0.545	0.083***	−0.015	(0.019)
Owns livestock	0.687	0.662	−0.025*	0.018	(0.017)
Land owned (hectares)	1.07	0.833	−0.238	−0.380	(0.306)
Rural	0.675	0.640	−0.035***	—	
Distance to major road (km)	8.34	3.75	−4.59***	—	
<i>Gestational climate</i>					
SPEI (6 month)	0.997	1.27	0.268***	−0.177**	(0.077)
NDVI z-score	0.678	0.747	0.070***	−0.164***	(0.034)
Observations	5,843	1,760			

Raw difference is the unconditional mean difference, exposed minus unexposed, with significance from a two-sample *t*-test. FE-adjusted is the coefficient on the exposure indicator from regressing each covariate on exposure with the same cluster and birth-cohort fixed effects, sampling weight, and grid-clustered standard error as column (3) of Table 2. ***, **, * denote significance at 1, 5 and 10 percent. Rural and distance to a major road are fixed within a DHS cluster and have no within-cluster variation once the cluster fixed effect is included, so no FE-adjusted coefficient is reported for either. Twin status and the birth interval to the next child likewise return no FE-adjusted coefficient in the same cluster-by-cohort structure, so none is reported for either. Counts in the observation block are for the full sample; the estimation sample for each covariate differs where a covariate has missing values (see the note on household head age above).

Table 10: Shifted-calendar 2014 pre-invasion falsification: cells significant at 5 percent

Exposure definition	Height-for-age z-score	Weight-for-age z-score	Weight-for- height z-score	Mortality
Proximity	0/9	1/9	0/9	0/9
+ Damaging form	0/9	0/9	0/9	0/9
+ Susceptible crop stage	0/9	0/9	0/9	0/9
+ Verified vegetation decline	0/9	0/9	0/9	3/9

Each cell counts how many of the nine radii (10 to 50 km) return a coefficient significant at the 5 percent level, out of 144 definition x outcome x radius cells, when the real invasion’s calendar pattern is shifted back 93 months onto the 2014 DHS cohort’s actual gestation windows (Section 5.3). 4 of 144 cells (2.8 percent) are significant, in line with the 5 percent nominal rate expected under no true pre-invasion effect. The three significant mortality cells are the verified measure at the 30, 35, and 50 km radii; the one significant weight-for-age cell is raw proximity at 50 km. No significant cell shares a definition, outcome, and adjacent radius with another, the pattern a genuine pre-existing trend would leave.

Table 11: Children flagged as exposed, by measure and radius

Radius (km)	Proximity	+ form	+ stage	+ verified	Disputed	Jaccard
10	2,249	2,140	1,320	728	1,521	0.32
15	2,889	2,792	1,798	1,067	1,822	0.37
20	3,349	3,263	2,148	1,333	2,016	0.40
25	3,721	3,618	2,431	1,529	2,192	0.41
30	4,070	3,991	2,728	1,760	2,310	0.43
35	4,350	4,256	2,962	1,911	2,439	0.44
40	4,628	4,526	3,155	2,046	2,582	0.44
45	4,833	4,727	3,351	2,164	2,669	0.45
50	5,031	4,904	3,503	2,285	2,746	0.45

Sample is 7,603 children. Measures nest, so no child is flagged by a stricter measure and not by a looser one. Disputed denotes children flagged by proximity and not by the verified measure. Jaccard is the overlap between the proximity and verified lists.

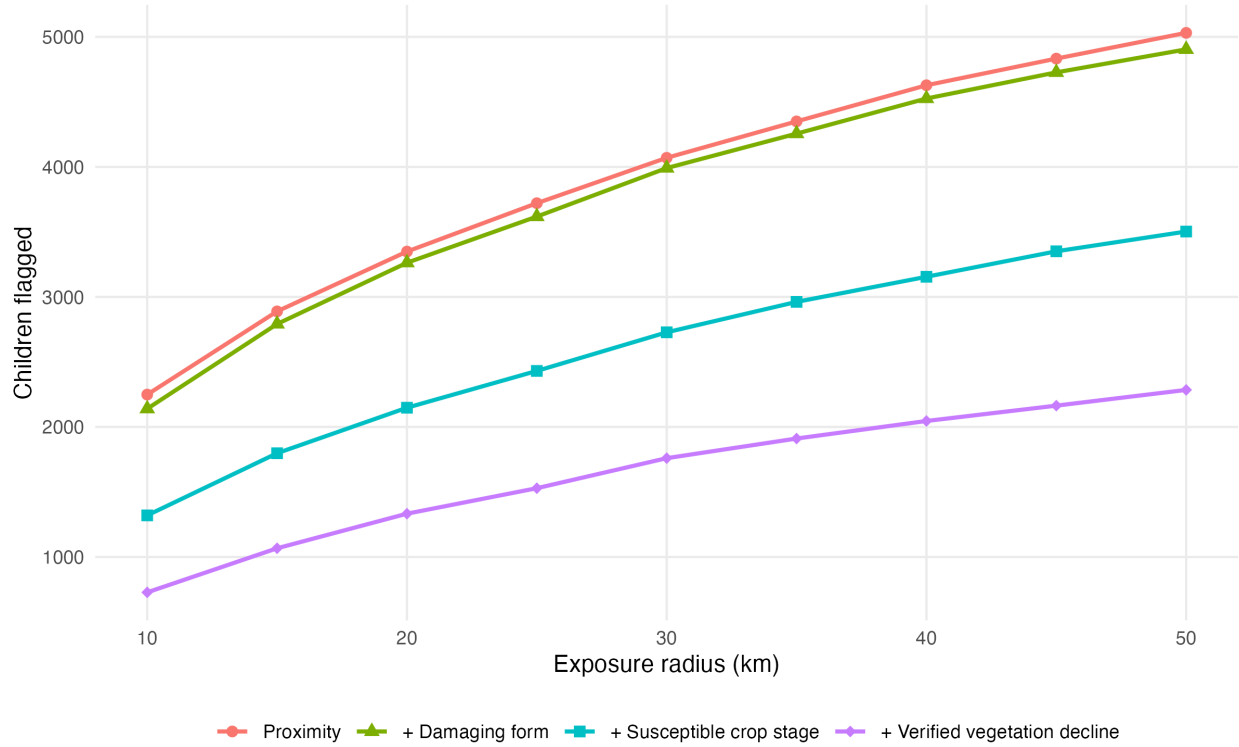


Figure 9: Children flagged as exposed, by exposure definition and radius. The four definitions nest, so every child flagged by a stricter measure is also flagged by every looser measure above it. Reproduces the counts of Table 11.

Table 12: Heterogeneity: baseline characteristics entered jointly

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score
Agricultural occupation	0.161 (0.117)	0.015 (0.132)	0.112 (0.128)
Owens land	0.036 (0.093)	−0.027 (0.097)	0.006 (0.096)
Rural	0.204* (0.111)	0.072 (0.124)	0.196* (0.112)
Distance to major road (km)	−0.000 (0.010)	0.016 (0.011)	0.008 (0.010)

Each column is a single regression of the column outcome on exposure at the main radius, all four baseline characteristics, their interactions with exposure, and the control set and fixed effects of column (3) of Table 2; each cell reports one characteristic’s exposure interaction from that shared regression, so the four rows in a column are directly comparable. Compare to the one-at-a-time version in the top panel of Table 5: rural residence is the only characteristic that keeps any significance once the other three are controlled for directly. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 13: Riverine-versus-cropping exposure gap, by radius

Radius (km)	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Treated in riverine
10	−0.197 (0.134)	−0.210 (0.153)	−0.231 (0.152)	23
15	−0.224 (0.136)	−0.218 (0.170)	−0.265* (0.146)	28
20	−0.269** (0.120)	−0.195 (0.160)	−0.288** (0.142)	28
25	−0.299** (0.120)	−0.203 (0.158)	−0.309** (0.142)	28
30	−0.249* (0.130)	−0.390** (0.191)	−0.372*** (0.132)	35
35	−0.294** (0.133)	−0.406** (0.189)	−0.413*** (0.133)	35
40	0.053 (0.161)	−0.057 (0.219)	−0.002 (0.179)	38
45	0.057 (0.159)	−0.056 (0.218)	−0.000 (0.178)	38
50	0.112 (0.177)	−0.047 (0.204)	0.039 (0.160)	39

Each row re-estimates the riverine-versus-cropping interaction from the bottom panel of Table 5 at the row’s radius in place of the main 30 km radius, with the same control set and fixed effects. The gap is negative at every radius, significant on at least one outcome from 15 through 35 km, the band the paper’s main dose-response results are built on, and falls to a small, sign-flipped, statistically indistinguishable-from-zero estimate at 40 km and beyond. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 14: Exposure and Infant mortality

Radius	Estimate	Treated	<i>p</i>
10 km	0.003 (0.009)	792	0.729
15 km	0.002 (0.008)	1,160	0.782
20 km	0.007 (0.007)	1,449	0.376
25 km	0.003 (0.007)	1,656	0.635
30 km	0.003 (0.006)	1,904	0.636
35 km	−0.000 (0.006)	2,066	0.944
40 km	0.005 (0.007)	2,209	0.434
45 km	0.004 (0.007)	2,338	0.509
50 km	0.004 (0.007)	2,467	0.600

Outcome is an indicator for death before interview. Standard errors clustered on a quarter-degree spatial grid in parentheses. Sample is all 8,213 estimable 2022 births rather than the measured subsample. Baseline mortality is 3.2 percent. Each row is a separate regression sharing the control set and fixed effects of column (3) of Table 2.

Table 15: Alternative Anthropometric Outcomes, 30 km radius

	Height- for-age z-score	Weight- for-age z-score	Weight- for-height z-score	Stunted	Underweight	Wasted
In utero exposure	−0.059 (0.071)	−0.167*** (0.064)	−0.185*** (0.055)	0.006 (0.024)	−0.002 (0.013)	0.019** (0.008)
Baseline prevalence				0.185	0.122	0.072
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cluster and birth-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,424	7,421	7,416	7,424	7,421	7,416
<i>R</i> ²	0.390	0.377	0.352	0.318	0.275	0.261

Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Each column is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Baseline prevalence is the share of unexposed children below the relevant threshold.

Table 16: In utero locust exposure and child stunting, 30 km radius

	Height-for-age z-score				Stunted	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.022 (0.071)	−0.035 (0.068)	−0.059 (0.071)	−0.071 (0.064)	0.006 (0.024)	0.021 (0.019)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,451	7,451	7,424	7,424	7,424	7,424
<i>R</i> ²	0.335	0.355	0.390	0.396	0.318	0.327

None of the six columns is significant at conventional levels. All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and birth intervals on either side. Family controls are mother's education, mother's height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a height-for-age z-score below −2, the clinical threshold for stunting.

Table 17: Dose response among verified-exposed children: number of damage events and height-for-age

Dependent variable: Height-for-age z-score	Linear	Log	Capped at 20
Verified damage events	−0.011 (0.009)		
Log(1 + verified damage events)		−0.182 (0.121)	
Verified damage events, capped at 20			−0.026 (0.016)
Observations	1,460	1,460	1,460

None of the three functional forms is significant at conventional levels. Standard errors clustered on a quarter degree spatial grid in parentheses. Sample restricted to children verified-exposed at the main 30 km radius. All columns include the control set and cluster by birth-cohort fixed effects of column (3) of Table 16. Controls for the number of RAMSES ground visits during gestation, counting every recorded visit within 30 km, of any outcome, during the child’s gestation window, to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

Table 18: In utero locust exposure and child underweight status, 30 km radius

	Weight-for-age z-score				Underweight	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.156** (0.063)	−0.167*** (0.061)	−0.167*** (0.064)	−0.189*** (0.062)	−0.002 (0.013)	0.015 (0.014)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,448	7,448	7,421	7,421	7,421	7,421
R^2	0.343	0.361	0.377	0.385	0.275	0.291

The point estimate is negative and significant in columns (1)-(4) but the binary (underweight) columns (5)-(6) are not significant, consistent with dilution similar to Section 5’s weight-for-age discussion at the main radius. All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and birth intervals on either side. Family controls are mother’s education, mother’s height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a weight-for-age z-score below −2, the clinical threshold for underweight.

Table 19: Dose response among verified-exposed children: number of damage events and weight-for-age

Dependent variable: Weight-for-age z-score	Linear	Log	Capped at 20
Verified damage events	−0.016** (0.008)		
Log(1 + verified damage events)		−0.280*** (0.089)	
Verified damage events, capped at 20			−0.037*** (0.012)
Observations	1,459	1,459	1,459

Standard errors clustered on a quarter degree spatial grid in parentheses. Sample restricted to children verified-exposed at the main 30 km radius. All columns include the control set and cluster by birth-cohort fixed effects of column (3) of Table 18. Controls for the number of RAMSES ground visits during gestation, counting every recorded visit within 30 km, of any outcome, during the child’s gestation window, to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

B Data and Descriptive Statistics

B.1 Demographic Health Survey Sample

Table 20 reports summary statistics for the 2022 DHS estimation sample. The anthropometric outcomes carry Kenya’s general burden. Mean height-for-age is -0.91 with a standard deviation of 1.31 , and a quarter of the sample sits below -1.77 . Weight-for-height is closer to the reference median at -0.21 , consistent with a population where chronic stunting is more prevalent than acute wasting. Children average 14.7 months at measurement and the sample is 49 percent female. Two thirds of households are rural, 48 percent own land, and 68 percent own livestock, but only 16 percent report an agricultural primary occupation, which is why the occupation variable enters only as a heterogeneity dimension and never as a control.

Distance to a major road is heavily right-skewed. The median cluster sits 2.2 km from one and the 75 th percentile 5.8 km, but the maximum exceeds 114 km. Gestational SPEI averages 0.80 at the three-month horizon and 1.06 at six months, both positive, reflecting that the invasion window overlapped a period of above-average moisture in much of the country. This is the same moisture that allowed the swarms to breed.

Table 20: Summary statistics, 2022 DHS estimation sample

	Obs.	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
<i>Anthropometry</i>								
Height-for-age (HAZ)	7,603	-0.91	1.31	-5.90	-1.77	-0.92	-0.11	5.95
Weight-for-age (WAZ)	7,600	-0.64	1.20	-5.29	-1.43	-0.63	0.13	4.85
Weight-for-height (WHZ)	7,595	-0.21	1.20	-5.00	-1.01	-0.20	0.58	4.82
<i>Child and mother</i>								
Age (months)	7,603	14.71	7.26	1.00	9.00	14.00	21.00	30.00
Female	7,603	0.49	0.50	0.00	0.00	0.00	1.00	1.00
First born	7,603	0.26	0.44	0.00	0.00	0.00	1.00	1.00
Twin	7,603	0.03	0.17	0.00	0.00	0.00	0.00	1.00
Birth order	7,603	3.23	2.24	1.00	1.00	3.00	4.00	14.00
Mother’s education (yr)	7,603	7.66	5.08	0.00	3.50	8.00	12.00	24.00
Mother’s height (cm)	7,603	160.76	4.44	101.70	160.40	160.70	161.00	180.40
<i>Household and location</i>								
Household size	7,603	5.90	2.57	2.00	4.00	5.00	7.00	24.00
Rural	7,603	0.67	0.47	0.00	0.00	1.00	1.00	1.00
Owns land	7,603	0.48	0.50	0.00	0.00	0.00	1.00	1.00
Owns livestock	7,603	0.68	0.47	0.00	0.00	1.00	1.00	1.00
Agricultural occupation	7,603	0.16	0.36	0.00	0.00	0.00	0.00	1.00
Distance to road (km)	7,603	7.28	15.65	0.00	0.74	2.18	5.82	114.65
<i>Gestational window</i>								
SPEI (3 month)	7,598	0.80	0.89	-1.75	0.08	0.91	1.58	2.42
SPEI (6 month)	7,598	1.06	1.07	-2.11	0.25	1.20	1.98	3.06
NDVI z-score	7,603	0.69	0.80	-3.04	0.24	0.72	1.19	6.04
Maize price index	7,603	1.11	0.15	0.81	1.03	1.09	1.14	1.90

Children in the 2022 DHS round whose cluster lies within 50 km of at least one locust monitoring visit.

Figure 10 maps the 2022 cluster locations against the major-road network and the KAMIS market directory used to interpolate prices. Clusters follow population density, dense through the western highlands, the central highlands, and the coastal strip, and sparse across the arid north and northeast. The road and market layers follow the same pattern, which is the source of the remoteness gradient that recurs throughout the results. The clusters farthest from a road are also the clusters whose price series rests most heavily on

interpolation and whose livelihood zones are least crop-dependent.

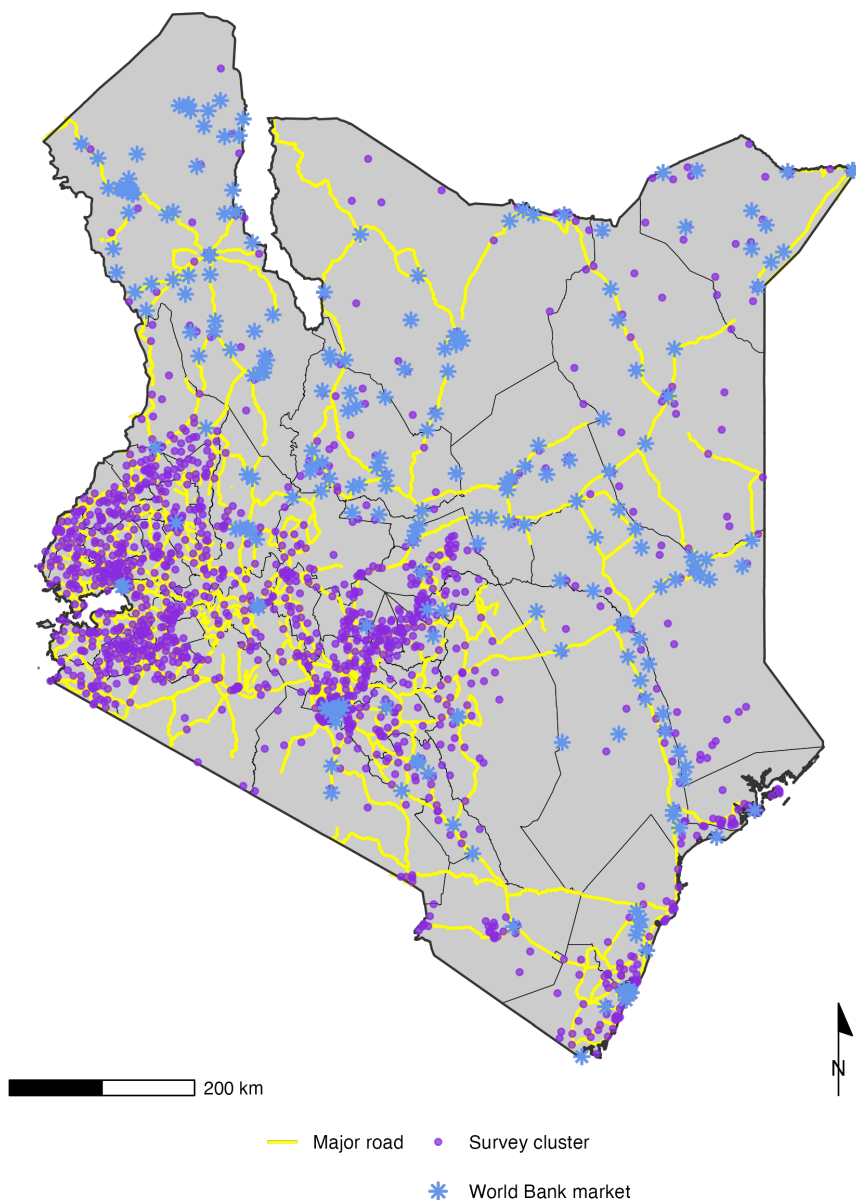


Figure 10: 2022 DHS cluster locations, major roads, and KAMIS market locations.

B.1.1 Sampling Weights

Figure 11 plots the distribution of DHS sampling weights on a log scale. The distribution is right-skewed, with the 99th percentile 6.3 times the median and the maximum 14.2 times, which motivates the winsorized robustness check in Section 5.3. Winsorizing weights at the first and 99th percentiles and re-running the full grid leaves the median estimate at -0.072 against -0.088 untrimmed, the negative share at 92 percent against 96 percent, and the significant share at 26 percent against 33 percent. Sample size and composition are identical under winsorizing, since weights are capped rather than observations dropped.

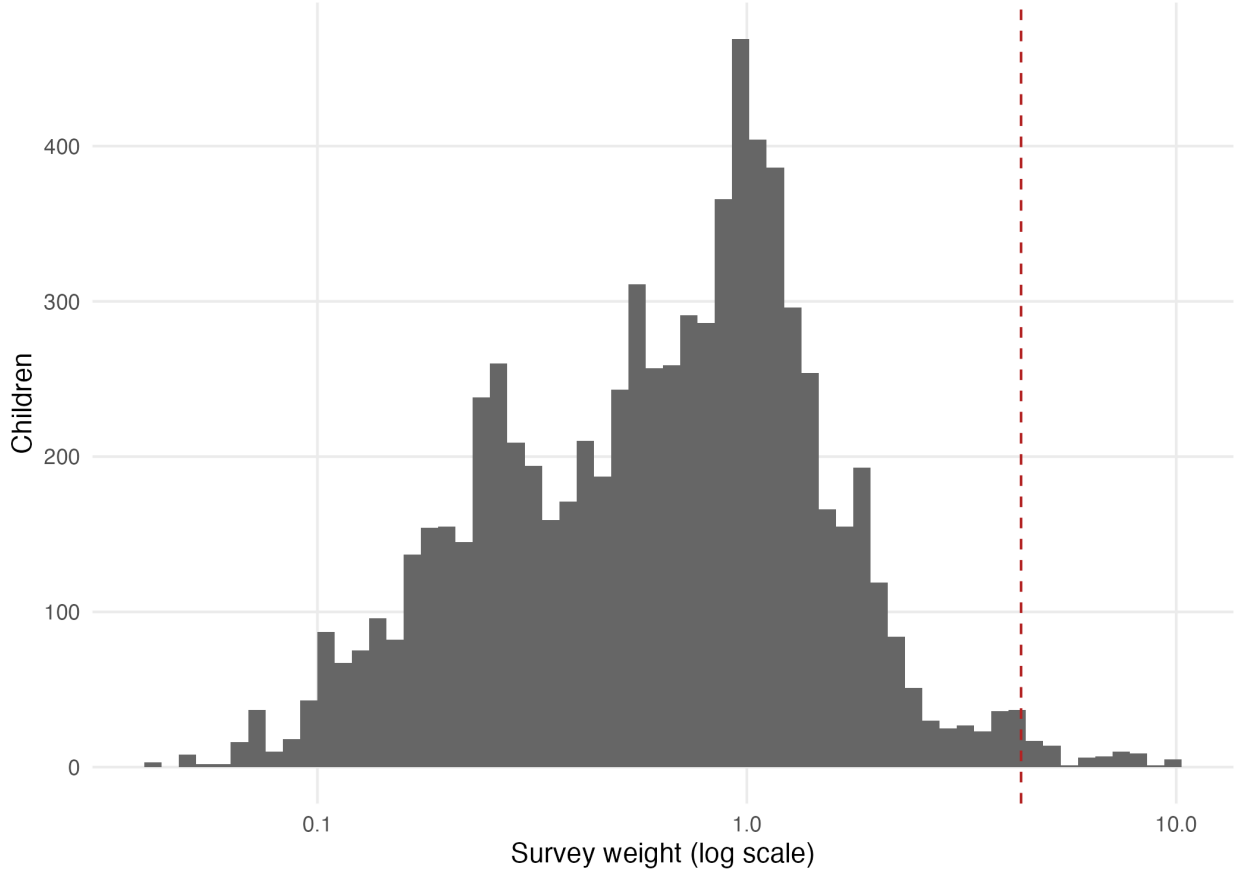


Figure 11: Distribution of DHS sampling weights. Dashed line marks the 99th percentile.

B.2 Climate and Vegetation Measures

Rainfall enters through CHIRPS at two temporal resolutions. The monthly product supports the gestational rainfall controls and the SPEI construction. The dekadal product, at ten-day resolution, supports growing-season onset detection. Onset cannot be identified from monthly totals, since a season that begins in the third dekad of March and one that begins in the first dekad of April differ by roughly ten days in the field but fall in adjacent calendar months, and the crop calendar shift described in Section 3.4 depends on resolving that difference. Extraction is a buffer mean over the DHS displacement radius rather than a point value.

SPEI is constructed at three- and six-month accumulation windows from CHIRPS precipitation and TerraClimate potential evapotranspiration, following [Vicente-Serrano et al. \(2010\)](#). The index standardizes the climatic water balance, precipitation minus PET, against that location’s own historical distribution, so a value of 1.0 means the gestational window was one standard deviation wetter than normal for that cluster rather than wetter than some national average. Only the six-month index enters the control set. The three- and six-month indices are highly correlated within cluster and including both adds collinear variation without adding information, so the longer window is retained as the one more likely to capture a full growing season.

Vegetation enters through MODIS MOD13Q1, a 16-day composite at 250 m resolution. Two features of this product govern how it is used. The compositing window means the date of an NDVI observation is not the date of the underlying satellite pass, since the product selects the best available pixel within each 16-day

period. This is why the pre- and post-arrival windows in equation (6) are defined in composites rather than in days, and why the post-arrival window spans roughly 16 to 48 days depending on where within its composite an arrival falls. Pixels also carry a reliability flag, and pixels flagged as cloud- or snow-contaminated are dropped before averaging to the cluster, so a composite with heavy cloud cover contributes fewer pixels rather than contributing corrupted ones.

NDVI is converted to a per-cluster, per-composite z-score against that cluster’s own pre-invasion climatology for the same composite of the year. This removes both the level difference between a humid highland cluster and an arid northern one and the seasonal cycle within each. A z-score of zero means vegetation was exactly at its normal level for that place at that point in the year. The damage condition in equation (8) then tests a change in this z-score rather than its level, so a cluster that was already dry relative to its own norm does not qualify on dryness alone.

B.3 Crop Calendars

Two sources fix the crop calendar, as Section 3.4 describes. This appendix reproduces an example of each, along with the transcribed calendar those sources are turned into for the crop this paper discusses most, maize.

Figure 12 shows the FEWS NET seasonal calendar for Livelihood Zone 19, the Central Highlands High Potential Zone, which covers the high-rainfall counties north of Nairobi. The zone runs a single extended long-rains season. Land preparation for the maize and bean crop occupies March and April, planting runs from late April into June, and harvesting begins in October and continues to January. Irish potato and cabbage run two cycles. Coffee and tea are harvested across most of the year, and a footnote lists permanent crops including avocado, mango, macadamia, banana, and pineapple. Food prices peak in May and June, in the gap between planting and harvest, and market access is poor during the rains. A swarm arriving in this zone in May or June meets a maize crop in its vegetative and reproductive stage, which is the highest susceptibility rank in equation (3). A swarm arriving in February meets bare ground.

Figure 13 shows the calendar for Livelihood Zone 9, the Northeastern Pastoral Zone, covering Wajir and the surrounding rangeland. The calendar has no crop section. It records camel calving, kidding and lambing, disease outbreaks, milk availability, migration, livestock sales, and the lean season. Households here obtain food from milk, livestock sales, and purchase, and the FEWS assessment calendar tracks livestock condition rather than crop condition. No date in this zone places a field in a susceptible stage, so equation (5) never holds and no child in this zone is treated. Section 5.4 reports what proximity exposure predicts in these zones, which is nothing.

Figure 14 collects the maize calendar transcribed from calendars like Figure 12, one row per livelihood zone. Zones without a maize row, such as the pastoral zone in Figure 13, are absent from the figure rather than shown blank. The land preparation and planting window moves with each zone’s rains, so the bimodal zones on the coast and around Lake Victoria show two separate cropping windows in a single calendar year while the single-season zones, including the Central Highlands zone above, show one.

Seasonal Calendar: Livelihood Zone 19 - Central Highlands, High Potential Zone

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Seasons	Dry		Long rains									Dry
Crops												
Legend	Land Preparation Planting Harvesting											
Maize + Beans												
Coffee												
Tea												
Irish potato												
Cabbage												
Pyrethrum	Include pruning											
Others*	Grown and harvested at different times throughout the year.											
Milk availability	Throughout the year due to dairy farming											
Livestock	Livestock Husbandry, vaccinations and general management throughout the year (intensive)											
Food prices					High							
Market access			Poor								Poor	
Labor			Peak									
Crop Assessments												
* Permanent crops including avocado, passion fruit, mangos, macadamia nuts, bananas and pineapples												

Figure 12: FEWS NET seasonal calendar, Livelihood Zone 19, Central Highlands High Potential Zone.

Seasonal Calendar: Livelihood Zone 9 - Northeastern Pastoral Zone

Seasonal Calendar: Livestock Zone 3 - Northeast Pastoral Zone	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Seasons	Dry			Long rains			Dry			Short rains		
Livestock production												
Camel Calving										Peak		
Kidding/lambing				Peak						Peak		
Disease outbreaks							Highly likely					
Prices				Peak			Peak					
Milk availability												
Camel milk												
Migration			Dry		Wet		Dry			Wet		
Others												
Livestock sales	High					High				High		High
Risk of insecurity				High				High			High	
Malnutrition										High		
Lean season												
Labor Availability			Peak								Peak	
Market access				Poor								Poor
Water stress												
Cross border inflows						Peak						
Food price					High					High		
FS Assessments												

Figure 13: FEWS NET seasonal calendar, Livelihood Zone 9, Northeastern Pastoral Zone. The calendar contains no crop section.

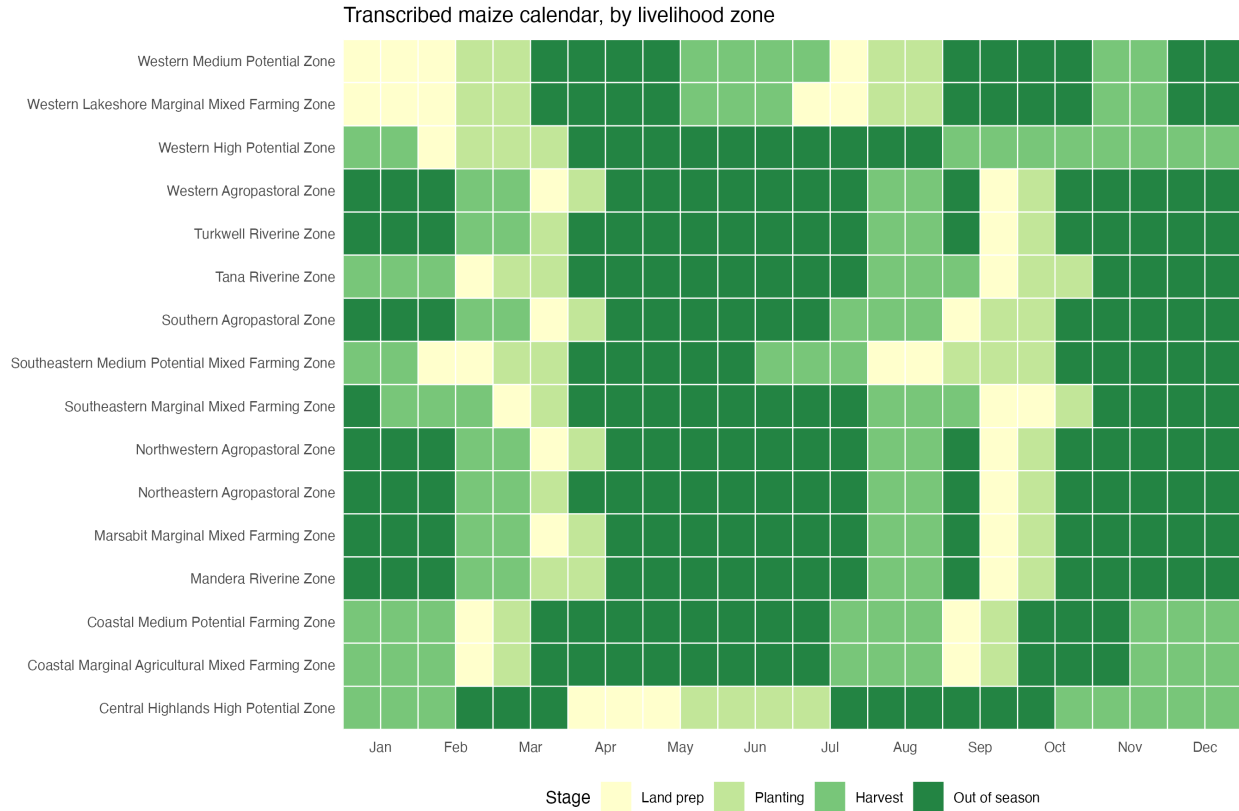


Figure 14: Transcribed maize calendar, by livelihood zone, built from calendars such as Figure 12.

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